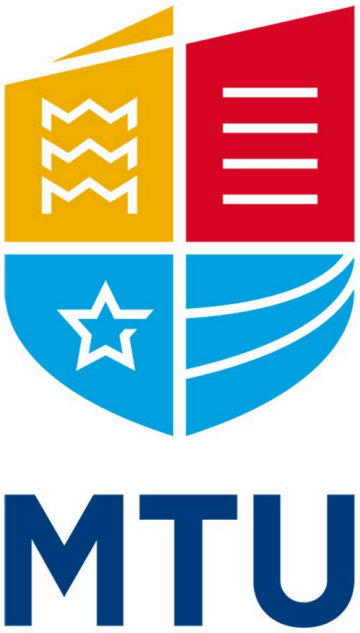


Multimodal Breast Cancer Detection

by

Betul Kantaroglu

This thesis has been submitted in partial fulfillment for the
degree of Bachelor of Science in Software Development

in the
Faculty of Engineering and Science
Department of Computer Science

May 2025

Declaration of Authorship

This report, Multimodal Breast Cancer Detection, is submitted in partial fulfillment of the requirements of Bachelor of Science in Software Development at Munster Technological University Cork. I, Betul Kantaroglu, declare that this thesis titled, Multimodal Breast Cancer Detection and the work represents substantially the result of my own work except where explicitly indicated in the text. This report may be freely copied and distributed provided the source is explicitly acknowledged. I confirm that:

- This work was done wholly or mainly while in candidature Bachelor of Science in Software Development at Munster Technological University Cork.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at Munster Technological University Cork or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this project report is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed: Betul Kantaroglu

Date: 11/05/2025

MUNSTER TECHNOLOGICAL UNIVERSITY CORK

Abstract

Faculty of Engineering and Science
Department of Computer Science

Bachelor of Science

by Betul Kantaroglu

Cancer is one of the leading causes of death worldwide [1]. Among many types of cancer, breast cancer remains one of the leading causes of cancer-related deaths worldwide. Early diagnosis is crucial in increasing survival rates. Although achieving rapid diagnosis is desirable, many limitations, such as data availability, complex mammograms, low resources, and an increase in workload on healthcare professionals, may hinder this.

This project aims to address these limitations by using AI and Machine Learning techniques on both unimodal and multimodal breast cancer datasets. Unimodal models (clinical metadata or mammograms) serve as baselines, allowing for comparisons against the multimodal models this project explores. The goal is to develop a robust AI-powered classification system that improves accuracy, reliability, and accessibility.

Multimodal models were trained with many different techniques, with transfer learning on both mammograms and clinical data serving as the baseline. Additionally, four ablation scenarios were tested to replicate missing data conditions: ablating images (clinical-only), ablating text (image-only), random modality drop (50%), and substitution of real images with synthetic GAN-generated data. Each experiment determined whether the model's diagnosing ability improved or decreased under restricted input conditions.

Results showed that the clinical-only model yielded the highest F1-score (0.962), followed by random modality drop and GAN-generated inputs with 0.93 and 0.96 F1-scores, respectively. Baseline multimodal model with transfer learning achieved an accuracy of 0.8148. Results show that clinical features alone are strong predictors, and synthetic data can improve robustness in low-resource scenarios.

In conclusion, this study confirms that multimodal models tested are a strong and viable tool for breast cancer detection.

Acknowledgements

I would like to express my sincere gratitude to my project supervisor, Jing Hua Ye, for her invaluable guidance, mentorship, support, and feedback throughout the duration of this project. Her insights and encouragement paved the way for the success of this project.

I would also like to thank Deirdre Dunlea, my second reader, for her time and useful insight during the evaluation stage of the project.

A special thanks to the Computer Science department and all the lecturers who provided me with the foundational knowledge and tools to complete this work during the course of this program.

I am deeply grateful to my mother, whose unwavering love, strength, and endless sacrifices have shaped every step of this journey.

Contents

Declaration of Authorship i

Abstract i

Acknowledgements iii

List of Figures viii

List of Tables x

Abbreviations xi

1 Introduction 1
- 1.1 Problem Statement . . . 1
- 1.2 Motivation . . . 2
- 1.3 Objectives . . . 2
- 1.4 Research Questions . . . 3
- 1.5 Contribution . . . 3
- 1.6 Structure of This Report . . . 4

2 Literature Background 6
- 2.1 Thematic Area within Computer Science . . . 6
- 2.2 Breast Cancer Overview . . . 9
  - 2.2.1 Common Symptoms . . . 9
  - 2.2.2 Traditional Diagnostic Methods (Without AI/ML) . . . 10
  - 2.2.3 Limitations of Traditional Methods . . . 12
- 2.3 Datasets . . . 12
  - 2.3.1 Clinical Datasets . . . 13
  - 2.3.2 Imaging Datasets . . . 13
  - 2.3.3 Histopathology Datasets . . . 13
  - 2.3.4 Genomic and Molecular Datasets . . . 14
  - 2.3.5 Challenges in Breast Cancer Datasets . . . 14
  - 2.3.6 Importance of Multimodal Datasets . . . 15
- 2.4 A Review of AI in Healthcare . . . 15
  - 2.4.1 Top 3 Most Relevant Texts/Resources to Be Used . . . 17

2.4.2 Top 5 Companies/Organizations Potentially Interested in This Project . . . 17
2.4.3 Top 5 Wiki/Forums/Blogs/YouTube Channels Related to This Topic 18
2.5 Literature Review of Multimodal Breast Cancer Detection and Associated Fields . . . 20
2.5.1 Medical Image-Based Breast Cancer Detection . . . 20
2.5.2 Clinical Data and NLP . . . 20
2.5.3 TL for Medical Images . . . 21
2.5.4 Synthetic Medical Images Using GANs . . . 21
2.5.5 Multimodal ML in Healthcare . . . 21
2.6 Trends Identified . . . 21
2.6.1 Multimodal ML Approaches . . . 21
2.6.2 Multimodal DL Approaches . . . 22
2.6.3 Modalities Ablation . . . 22
2.6.4 Multimodal ML Methods with Missing Modalities . . . 22
2.6.5 Multimodal DL Techniques with Missing Modalities . . . 23
2.6.6 TL . . . 23
2.6.7 Generative AI Techniques . . . 24

3 Problem -Multimodal Breast Cancer Detection with Deep Learning and GANs 25
3.1 Functional Requirements . . . 25
3.2 Non-Functional Requirements . . . 26
3.3 Project Plan . . . 27
3.3.1 Technologies . . . 27
3.3.2 How the System Works . . . 28
3.3.3 Hardware and Tools . . . 28
3.4 Project Risks and Mitigation Strategies . . . 28
3.4.1 Shortage of Labeled Data for DL and GANs . . . 29
3.4.2 Multimodal Data Incompatibility in Fusion . . . 29
3.4.3 Model Overfitting due to Small Datasets . . . 29
3.4.4 Unnatural or Unreliable Text Output from LLMs . . . 29
3.4.5 Failure of GANs to Generate Useful Synthetic Images . . . 30
3.4.6 Difficulty Interpreting Model Predictions . . . 30
3.4.7 Loss of Progress due to Poor File Management or No Version Control 30
3.4.8 Time Management and Assignment Overlap . . . 30
3.4.9 Misalignment with Supervisor Expectations . . . 30
3.4.10 Critical Risk Action Plan . . . 31

4 Architecture and Project Design 33
4.1 Overall Architecture . . . 33
4.1.1 Modality A: Image Data Pipeline . . . 33
4.1.2 Modality B: Clinical Data Pipeline . . . 34
4.1.3 Modality B: Unimodal Models . . . 34
Architecture: . . . 34
4.1.4 GAN-Synthesized Image Branch . . . 35
4.1.5 Modality Ablation and Replacement . . . 35

4.1.6 Feature Fusion and Classification . . . 35
4.1.7 System Flexibility and Experimental Configurations . . . 35
4.1.8 Summary . . . 36
4.2 Methodology . . . 36
4.2.1 Background Research Strategy . . . 36
4.2.2 Technical Setup and Skill Development . . . 37
4.3 Evaluation . . . 37
4.3.1 Evaluation Goals . . . 37
4.3.2 Evaluation Metrics . . . 38

5 Experimental Setup and Model Designs 40
5.1 Datasets and Preprocessing . . . 40
5.1.1 Datasets . . . 40
5.1.2 Unimodal Dataset Preprocessing . . . 41
5.1.3 MultiModal Dataset Preprocessing . . . 42
5.1.4 Sample Size and Class Balance . . . 43
5.1.5 Data Modalities . . . 43
5.1.6 Controlled Environment and Collection Conditions . . . 44
5.2 UNIMODAL DATASETS: General Experimental Settings . . . 44
5.2.1 Environment . . . 44
5.2.2 Training/Validation/Test Split Ratios . . . 45
5.2.3 Structured Unimodal Model Hyperparameters . . . 45
5.2.4 Validation Methods and Loss Functions . . . 46
5.2.5 Number of Iterations/Epochs and Training Procedure . . . 46
5.3 AISSLAB - MULTIMODAL: General Experimental Settings . . . 47
5.3.1 Environment . . . 47
5.3.2 Training/Validation/Test Split Ratios . . . 47
5.3.3 Multimodal Model Hyperparameters . . . 48
5.3.4 Validation Methods and Loss Functions . . . 48
5.3.5 Number of Iterations/Epochs and Training Procedure . . . 49
5.4 Unimodal Experiments - AISSLAB . . . 50
5.4.1 Benchmark Replication Approach . . . 50
5.4.2 Models Used and Architectural Details . . . 50
5.5 Multimodal Experiments without Ablations . . . 53
5.5.1 Overall Architectural Scheme . . . 53
5.5.2 Visual Feature Extraction (CNN Layer Details) . . . 53
5.5.3 Text/Other Modality Feature Extraction (RNN/Transformer Layer Details) . . . 54
5.5.4 Cross-Modal Fusion Methods . . . 55
5.6 Multimodal Experiments with Transfer Learning . . . 55
5.6.1 Selection of Pre-trained Models . . . 55
5.6.2 Transfer Strategies and Fine-tuning . . . 57
5.7 Multimodal Experiments with Ablation Studies . . . 58
5.7.1 Ablation Design and Objectives . . . 58
5.7.2 Component-wise Isolated Tests . . . 59
5.8 GAN-based Multimodal Experiments . . . 59
5.8.1 GAN Architecture and Components . . . 59

5.8.2 Training Procedure and Loss Functions . . . 60
5.8.3 Training Procedure and Loss Functions . . . 61

6 Evaluation and Results 62
6.1 Unimodal Baselines - Evaluation and Results . . . 62
6.1.1 WDBC Dataset . . . 62
Confusion Matrices: . . . 62
Classification Metrics: . . . 64
ROC Curve: . . . 65
Conclusion: . . . 65
6.1.2 WBC Dataset . . . 65
Confusion Matrix: . . . 66
Classification Metrics: . . . 66
ROC Curve and AUC: . . . 67
Conclusion: . . . 67
6.1.3 BCSC Dataset . . . 68
1. Logistic Regression: . . . 69
2. Random Forest: . . . 70
3. XGBoost: . . . 71
ROC Curve: . . . 72
Conclusion: . . . 73
6.1.4 Summary & Analysis of Unimodal Results . . . 73
6.2 Multimodal results . . . 74
6.2.1 GAN-Based Ablation . . . 74
6.2.2 Random-Drop Fusion Ablation . . . 75
6.2.3 Text-Only Ablation . . . 75
6.3 Comparative Training Performance . . . 76
6.4 Final Metrics Comparison . . . 77
6.5 TL Model Evaluation . . . 78
6.5.1 Training and Validation History . . . 78
6.5.2 Confusion Matrix . . . 79
6.5.3 ROC Curve . . . 79
6.6 Overall Comparison and Final Remarks . . . 80

7 Conclusions And Future Work 84
7.1 Conclusions . . . 84
7.2 Future Work . . . 86

Bibliography 88

A Appendix: Poster vs Report Results 92
A.1 Result Discrepancies Note . . . 92

List of Figures

2.1 The visualization of breast cancer - benign and malignant tumor cells and masses 9

2.2 Breast Self-Examination 11

4.1 System architecture indicating the processing pipelines of every modality (image, clinical, synthetic), TL, modality ablation, GAN-based image synthesis, and ultimate feature fusion and classification. 34

5.1 The neuron architecture for the CNN in unimodal detection 51

5.2 The overall architecture of CNN in unimodal detection 52

5.3 Feature Pyramid Network 54

5.4 Tansfer Learning with EfficientNet-B0 56

5.5 DenseNet Architecture 56

5.6 Transfer Learning Pipeline With EfficientNet-B0 57

5.7 Multimodal Ablation Study Architecture 58

5.8 GAN architecture for ablation study 60

6.1 Confusion Matrix: Logistic Regression on WDBC Test Set 64

6.2 Confusion Matrix: MLP Neural Network on WDBC Test Set 64

6.3 ROC Curve for WDBC Dataset: Logistic Regression vs. MLP Neural Network 65

6.4 Confusion Matrix for Logistic Regression on WBC Test Set 66

6.5 ROC Curve for Logistic Regression on WBC Test Set (AUC = 0.99) . . . 67

6.6 Balanced Class Distribution After SMOTE 69

6.7 Confusion Matrix - Logistic Regression on BCSC 70

6.8 Confusion Matrix - Random Forest on BCSC 71

6.9 Confusion Matrix - XGBoost on BCSC 72

6.10 ROC Curve Comparison for BCSC Models 73

6.11 Training and validation loss and F1 score for GAN-only model 74

6.12 Randomly remove one modality 75

6.13 Performance of models with fully ablating text only 76

6.14 The comparative training and validation performance 76

6.15 The summary of each ablation's key test metrics, accuracy, precision, recall, and F1 77

6.16 TL training validation results 78

6.17 The confusion matrix for TL (test set N = 60) 79

6.18 The ROC curve for the TL model 80

6.19 The final test metrics for all the models and ablation variants 81

List of Tables

2.1 PET Physical Properties 20

3.1 Key Risks and Mitigation Strategies 32

6.1 Performance Summary of Unimodal Models Across All Datasets 74

Abbreviations

AI	**A**rtificial **I**ntelligence
ML	**M**achine **L**earning
DL	**D**eep **L**earning
SVM	**S**upport **V**ector **M**achine
LLM	**L**arge **L**anguage **M**odel
NLP	**N**atural **L**anguage **P**rocessing
OOP	**O**bject-**O**riented **P**rogramming
EDA	**E**xploratory **D**ata **A**nalysis
CBE	**C**linical **B**reast **E**xamination
BCRAT	**B**reast **C**ancer **R**isk **A**ssessment **T**ool
BCSC	**B**reast **C**ancer **S**urveillance **C**onsortium
WBC	**W**isconsin **B**reast **C**ancer
WDBC	**W**isconsin **D**iagnostic **B**reast **C**ancer
FNA	**F**ine **N**eedle **A**spirate
DDSM	**D**igital **D**atabase for **S**creening **M**ammography
FNN	**F**eed-**F**orward **N**eural **N**etwork
CNN	**C**onvolutional **N**eural **N**etwork
GAN	**G**enerative **A**dversarial **N**etwork
SHAP	**Sh**apley **A**dditive **E**xplanation
AUC	**A**rea **U**nder the **C**urve
ROC	**R**eceiver **O**perating **C**haracteristic
FPR	**F**alse **P**ositive **R**ate
TL	**T**ransfer **L**earning
MDCMI-BC	**M**a'amon's **D**iagnostic **C**entre **M**ammogram **I**mages for **B**reast **C**ancer
CC	**c**ranio**c**audal

MLO	**m**edio**l**ateral **o**blique
DCGAN	**D**eep **C**onvolutional **GAN**

This work is dedicated to my mother, who never stopped believing in me...

Chapter 1

Introduction

In this chapter, the problem being solved and the main motivation behind this project will be discussed. Moreover, the main research questions and contributions to this project will also be outlined.

1.1 Problem Statement

Breast cancer is a disease in which abnormal breast cells grow out of control and form tumours. These tumours can spread throughout the body and can result in fatality if left untreated. Breast cancer remains one of the most common cancers among women worldwide and contributes significantly to cancer-related deaths. According to the World Health Organisation (WHO), breast cancer took the lives of 670,000 patients in 2022 alone [2]. In Ireland, 3,600 women and approximately 30 men are diagnosed with breast cancer each year [3]. Breast cancer takes the lives of roughly 753 women and 8 men ANNUALLY [4]. Early detection and accurate classification are crucial for improving patient outcomes, allowing for timely treatment and lowering mortality rates. Breast cancer survival rates are increasing, and breast cancer accounts for 22% of cancer survivors. [4].

With the rise of digital healthcare and access to large datasets, machine learning (ML) and deep learning (DL) have shown a promising future in terms of assisting breast cancer diagnosis.

However, traditional diagnosis relies on single-modality information, such as mammographic images or clinical features, limiting the scope and accuracy of prediction. In real clinical settings, patients typically possess incomplete or noisy data — e.g., missing imaging reports or medical records. This project tries to overcome that limitation by

multimodal deep learning, integrating image, clinical, and text data to build a flexible, robust diagnostic model.

1.2 Motivation

The motivation behind this project lies in the potential of combining multiple data modalities, such as clinical records, imaging, and textual information, to enhance diagnostic accuracy and model robustness. Although previous research has produced strong results using individual data types, such as mammograms or clinical variables, combining different sources of information can give a more complete picture of a patient's condition.

On a personal and academic level, this project has been a valuable opportunity to apply skills from various fields of computer science, including natural language processing, computer vision, and traditional Machine Learning. It also allowed tackling a meaningful real-world healthcare problem using artificial intelligence (AI), an area of growing importance in both research and practice.

1.3 Objectives

The general goal of this project is to explore whether the mixture of synthetic and real inputs across modalities improves model accuracy, resistance to data missingness, and practical use in clinical diagnostic pipelines. By employing ablation studies (removing some modalities), synthetic data generation through Generative Adversarial Networks (GANs), and text simulation through Large Language Models (LLMs) like BioBERT and ClinicalBERT, the project will compare multiple configurations to juxtapose trade-offs and strengths.

In technical terms, the classification function can be defined as:

$$f(x) = f(x_c, x_i, x_t) \rightarrow y \quad \text{where} \quad y \in \{0, 1\}$$

where x is the multimodal input vector composed of clinical attributes x_c, imaging features x_i, and text embeddings x_t. The function f maps the multimodal input to a binary output y, representing a benign (0) or malignant (1) diagnosis.

To achieve this, the project will create a classification system that will be capable of working on different types of data and assess how it performs under different conditions.

For example, the system will be tested when some types of data are missing or replaced with artificial examples. These tests will examine how accurate the model is and how it might respond in actual clinical settings where patient data is not always present.

The majority of the project is comparing a few versions of the system, one with all data, and some with one element missing or replaced. It's more about viewing how each piece of data impacts the end product than creating one working model. The end system should provide some sense of:

- How text, images, and clinical data assist in enhancing diagnosis.
- To what extent does each data class impact performance?
- To what extent synthetic data may be replaced with actual data without impacting heavily on performance.
- To what extent pre-trained image models (e.g., EfficientNet) can be enhanced in sparse training data settings.

1.4 Research Questions

In this project, the following questions aim to be answered:

1. Can combining image and textual data improve breast cancer classification accuracy compared to an unimodal approach?
2. How does the ablation of one modality affect the performance of the model?
3. Can GAN-generated synthetic images be used to fill in the missing real medical images without significantly reducing accuracy?
4. What is the impact of transfer learning (TL) in multimodal medical classification tasks?

1.5 Contribution

This project focuses on breast cancer classification using either a multimodal DL approach or a multimodal ML approach. It integrates clinical data, imaging features, and simulated textual patient history. The main contributions are as follows:

Model Implementation: A variety of models were implemented and tested, including logistic regression, support vector machine (SVM), EfficientNet for visual input, and transformer-based models for text. TL techniques were used to improve generalization, and performance reached up to 97% accuracy on balanced datasets.

Ablation Studies: To evaluate the robustness of the multimodal pipeline, two ablation strategies were applied:

Random modality ablation: Randomly removing one or more data types to test how model performance changes.

Synthetic modality refillment: Refill those missing real images with GAN-generated ones to observe the effect of degraded or artificial inputs.

Evaluation and Benchmarking: Performance was compared to existing studies, such as Agarap's work on the WDBC dataset [5]. Metrics, including precision, recall, and F1-score, were calculated, with a particular focus on reducing false negatives, crucial in cancer diagnostics.

Tooling and Deployment: The project involved various preprocessing tasks like normalization, label encoding, and dataset merging. It also used Python libraries such as Scikit-learn, TensorFlow/Keras, and HuggingFace Transformers for model development and evaluation.

This project applied a wide range of topics from the computer science curriculum, including data preprocessing, ML workflows, evaluation strategies, multimodal integration, and TL. This project will contribute to the continuous evolution of intelligent diagnosis systems in medicine and offering a scalable and robust system that can be used in other medical classification problems aside from breast cancer.

1.6 Structure of This Report

The structure of this report will be as follows:

Introduction: General information about the project is discussed thoroughly, along with the motivation behind the project and its respective contributions.

Background: An extensive review of the techniques and approaches carried out in this field is conducted, and relevant sources and resources are discussed.

Problem: The problem to be solved is discussed.

Implementation Approach: The architecture of the framework and its methodology are explained, along with associated risk factors.

Research Phase Conclusions and Future Work: Based on the research, what conclusions were reached and what future work can be conducted are explored.

Experimental Setup and Model Designs - Implementation: The implementation phase is discussed, including how it was conducted, the methods, techniques, and approaches used.

Evaluation and Results: The results from the experiments are evaluated.

Conclusions And Discussions: Conclusions from the implementation phase of the project and its results are discussed. Future work is explored.

Chapter 2

Literature Background

This chapter aims to explore the place of this project within Computer Science.

2.1 Thematic Area within Computer Science

The project falls under the broader theme of AI in Healthcare, specifically multimodal learning, ML, and Natural Language Processing (NLP). It aims to improve the accuracy and robustness of breast cancer classification by combining various data sources, including structured clinical data and diagnostic images. Additionally, the project explores how an LLM can assist in interpretability during post-analysis, offering insights into classification decisions and potential clinical relevance.

In actual clinical practice, information is rarely perfect; not every patient may have a scan, scans may be missing for other reasons, patients may have incomplete histories, or unclear records. The aim is to be able to diagnose patients accurately, regardless of these factors. This project tries to simulate the explained real-world imperfections with modality ablation (e.g., removing scans, removing clinical text, or replacing the mammograms with GAN-generated images) to quantify the system's robustness. This also explores the possibility of synthetic augmentation in healthcare, especially in resource-constrained settings.

The key goal of the project is multimodal breast cancer classification using AI and ML techniques. By leveraging various data modalities—numerical attributes, visual data (mammographic images), and structured clinical text—the model can learn more intricate patterns and potentially achieve higher accuracy than unimodal models. Additionally, an LLM was explored during post-analysis to assist with interpretability and support clinical insight generation.

The project consists of core fields of computer science, such as:

ML and DL: For model building, training, and model selection, like logistic regression, SVM, EfficientNet, and transformers.

NLP: This will be explored during post-analysis using LLMs such as BioBERT and ClinicalBERT, helping to interpret classification outputs and extract clinical insights from structured text data.

Computer Vision: For image data management, including the use of a convolutional neural network (CNN) and TL via EfficientNet and ResNet-50.

Data Fusion and Integration: Merging clinical, image, and text data into a model pipeline.

Medical Informatics: Applying technical methods to solve healthcare issues using structured and unstructured health data.

Generative AI: Synthetic mammogram images for each patient, in response to their existing data, are implemented through GANs. This assists in addressing data scarcity and improving robustness during ablation testing.

Data Fusion and Integration: Techniques utilised for merging multimodal data (clinical metadata, images, and synthetic inputs) into unified model pipelines. This included early fusion strategies (feature concatenation) and ablation experiments to assess reliability under missing modality conditions.

Relevance of This Project to Software Development Modules at MTU Cork This project is especially relevant to some of the modules taught under the MT800 Software Development course, as it brings together a range of computer science and programming concepts, from programming fundamentals to ML and advanced data analytics. Implementing this knowledge in a real-world healthcare setting in such an integrated way demonstrates the comprehensive understanding and integration of such academic lessons.

SOFT6018 – Programming Fundamentals: This module laid down core programming concepts, like data types, variables, loops, conditionals, and functions. These were essential in laying down the machinery of the ML models and data preprocessing steps within the project. Writing clean, readable, and module-based code within the early design of the system architecture and experiments reflected a high degree of foundation in programming, especially when it involved handling different scripts for text preprocessing, data manipulation, and experimentation.

SOFT6017 – Modular Programming: Modular Programming focuses on organizing code into reusable, testable modules. This idea was crucial in arranging the different subsystems of this project — image preprocessing, GAN generation, model training, evaluation pipelines, and data fusion routines. Each modality (image, clinical, text) had different preprocessing and modeling steps, implemented as independent modules that could be reused or modified for ablation studies and testing. This module influenced the modular structure of the overall pipeline.

Object-Oriented Programming (in SOFT7011 or SOFT7023): Object-Oriented Programming (OOP) principles such as encapsulation, inheritance, and abstraction were applied across the system. For example, creating class-based wrappers for models, dataloaders, and evaluation scripts facilitated scalability and easier experimentation. OOP was very handy when creating the abstraction of model variants (e.g., text-only, image-only, multimodal) so that the implementation was cleaner and more receptive to change, such as switching models or datasets.

Programming for Data Analytics (SOFT8032): This module covers Python programming for data analysis of big data and visualization of outcomes. It played a vital role in this project: Exploratory Data Analysis (EDA) of clinical and imaging data, Normalisation and transformation pipelines, Merging datasets for multimodal learning, Visualization of training/testing outcomes (e.g., plotting accuracy and loss using matplotlib/seaborn), Building performance tables and figures for TL and ablation outcomes. The skills applied in this module enabled critical judgment in model assessment and result interpretation.

Machine Learning (COMP8043): This module has a direct application to the project's nature — developing models to classify breast cancer with both unimodal and multimodal inputs. It instructed:

- Supervised learning techniques such as logistic regression and SVMs, which were employed as baseline models.
- Evaluation metrics such as accuracy, precision, recall, and F1-score, all of which are commonly applied in this project.
- Principles for model training and testing, upon which the usage of cross-validation, early stopping, and benchmarking performance has been guided.
- Theories behind TL, specifically applied to reusing EfficientNet for image classification issues.
- And to top that off, theory in model resilience through ablation, and simulated data through GANs, conveys even more advanced motifs most commonly used to discuss topics under real-world problems of deploying models, each related through the emphasis within this module's interest in practical ML.

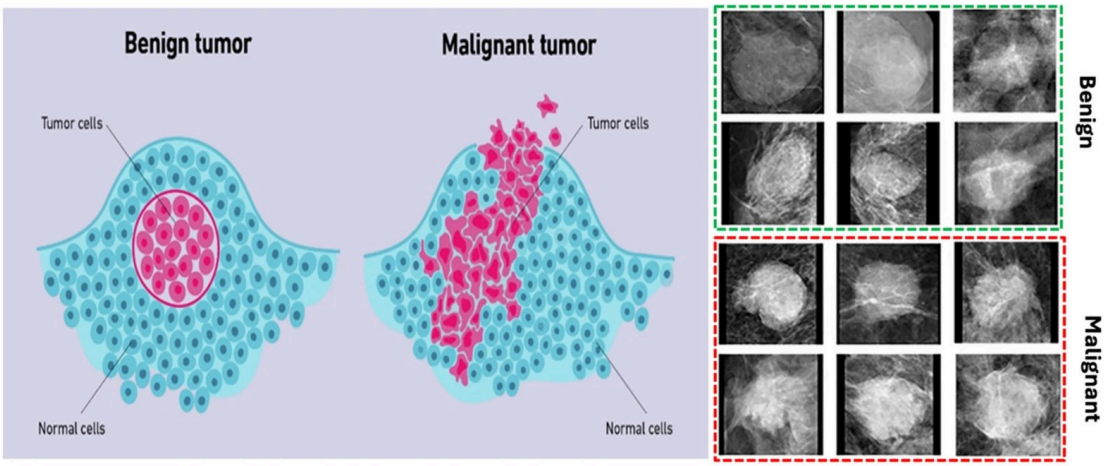


FIGURE 2.1: The visualization of breast cancer - benign and malignant tumor cells and masses

This is a real-world application of the software development curriculum, integrating programming methodologies, analytics, and ML in a practical context. Each module has its role in contributing to the project's planning and implementation.

2.2 Breast Cancer Overview

Breast cancer is one of the most common forms of cancer affecting women worldwide. It occurs when abnormal cells in the breast tissue begin developing and growing at an uncontrollable rate, forming tumours. In many cases of late diagnosis, the tumour may invade surrounding tissues or organs. Most commonly, it develops in the breast tissue, often in the ducts (which are channels responsible for carrying milk from lobules to the nipple) or in the lobules themselves (glands that produce milk). While breast cancer primarily affects women, it is important to note that breast cancer can also be diagnosed in men, in rare cases.

Breast cancer is a heterogeneous disease (see Fig. 2.1), meaning it presents in various subtypes, each with different biological characteristics, responses to treatment, and outcomes. Early detection plays a vital role in increasing the chances of successful treatment, to reduce the risk of spreading to other parts of the body, hence why a major emphasis on screening programs, regular checkups, and advances in diagnostic technology is encouraged.

2.2.1 Common Symptoms

The most common symptoms of breast cancer include:

- A sudden lump or thickening in or near the breast or underarm area, which may feel different from the tissue surrounding it.
- Changes in the size, appearance, and shape of the breast. This may be sudden or gradual.
- Unusual discharge from the nipple, with or without squeeze, may be in a clear form or may have a color.
- Redness, scaling, or dimpling of the skin on the breast, often resembling the peel of an orange (referred to as peau d'orange).
- An inverted nipple or any other visible changes in the appearance of the usual breast.

It is important to note that these symptoms do not always indicate cancer; however, any changes should be immediately evaluated by a qualified healthcare professional. Early diagnosis and treatment save lives.

Breast Self-Examination (BSE) does not replace the evaluation of a medical professional; however, it can provide individuals a sense of familiarity with the normal look and feel of their breasts. Performing regular self-examinations can help catch any changes that may have occurred (lumps, skin dimpling, skin scaling, discharge from the nipple, changes in size or shape). BSE is recommended to be performed once a month, ideally at the end of the menstrual cycle (for women), when breasts are least likely to be swollen or tender.

2.2.2 Traditional Diagnostic Methods (Without AI/ML)

Before the introduction of AI and advanced ML methods, breast cancer diagnosis relied almost entirely on manual processes and clinical expertise. The traditional way of diagnosis includes the following key methods:

Clinical Breast Examination (CBE): A clinician physically examines the breasts, checking for abnormalities such as lumps, swelling, or skin changes. This is often the first step, especially if the patient reports symptoms or during routine health checks.

Mammography: Mammography is a specialized X-ray technique used to capture detailed images of breast tissue. It is widely used for screening and early detection. Radiologists manually analyze the images to look for signs of cancer, such as masses

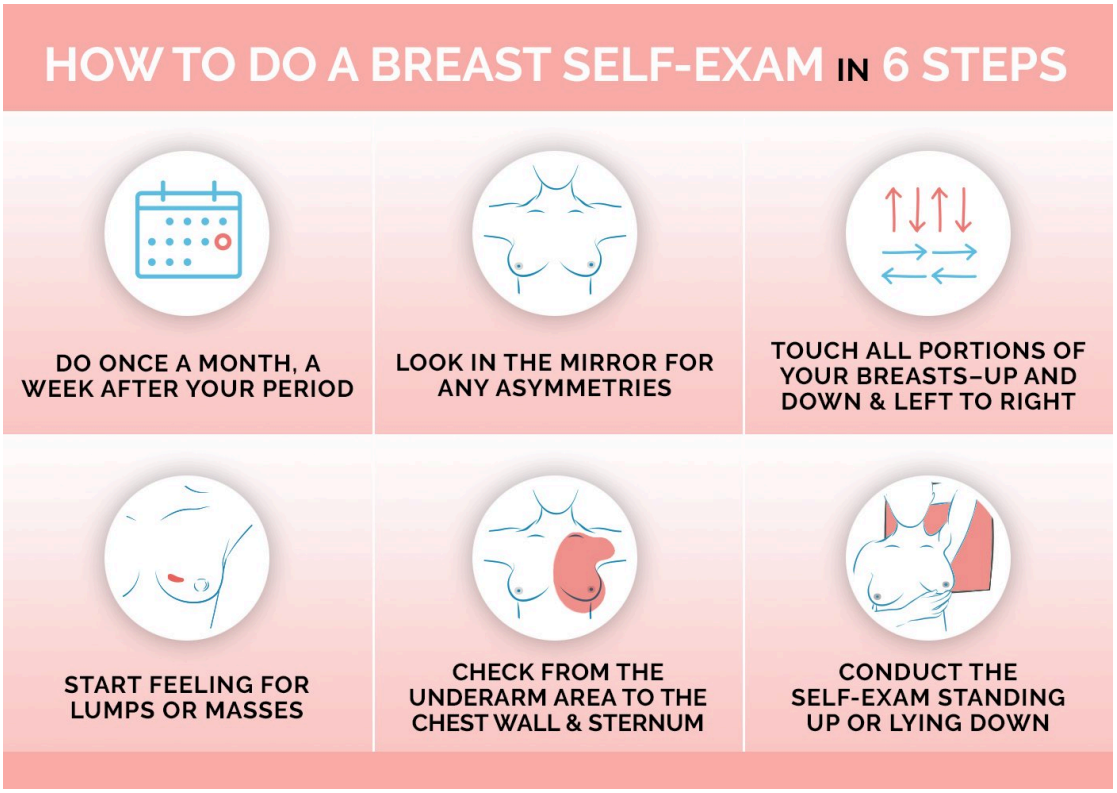


FIGURE 2.2: Breast Self-Examination

or calcifications. However, mammogram accuracy can vary, especially in patients with dense breast tissue.

Ultrasound: Ultrasound uses high-frequency sound waves to create images of the inside of the breast. It is often used as a follow-up test to clarify abnormalities found in a mammogram or physical exam, especially to distinguish between solid tumors and fluid-filled cysts.

Biopsy: When imaging or examination results are suspicious, a biopsy is performed. This involves removing a small piece of tissue from the breast for lab analysis. Biopsies can be done using a fine needle, core needle, or surgical method. The sample is analyzed by a pathologist to determine if it is benign or cancerous.

Histopathological Examination: Once a biopsy is taken, a more detailed analysis is performed under a microscope. The pathologist looks at the structure and characteristics of the cells, which can reveal information about the type, grade, and aggressiveness of the cancer, as well as hormone receptor and HER2 status.

Risk Assessment Models: Traditional tools like the Gail Model or the Breast Cancer Risk Assessment Tool (BCRAT) [6] are used to estimate an individual's risk of developing breast cancer. These tools consider factors such as age, family history,

personal health history, and reproductive factors. While useful, they are statistical and not tailored to individual images or real-time clinical data.

2.2.3 Limitations of Traditional Methods

While traditional diagnostic methods have saved countless lives and are still used widely, they have several limitations:

Human Error: Interpretation of mammograms or ultrasound images can vary between radiologists, leading to inconsistent results and causing confusion and stress to patients.

False Positives/Negatives: At times, dense breast tissue or subtle abnormalities can confuse and hence result in misdiagnosis.

Time-Consuming: Even in ideal conditions, manual diagnosis and pathology processes are often a lengthy and tiring process; this process can be even more tiring and even more delayed for individuals, clinics, and countries with low or limited resources.

Data Overload: Advancement in imaging technologies and the increase in patient data can create a strain on health care professionals as they are faced with a large volume of data to interpret, resulting possibly in cognitive fatigue, oversight, and reduced diagnostic efficiency.

These challenges highlight the importance of integrating newer technologies such as AI and multimodal ML into breast cancer diagnosis. These tools provide support to healthcare professionals, particularly in settings where expert radiologists or pathologists may not be readily available, enhancing accuracy, speed, and consistency.

2.3 Datasets

Breast cancer research has greatly benefited from the availability of publicly accessible datasets that enable the development, testing, and benchmarking of ML and DL models. These datasets vary in terms of modality (clinical, imaging, histopathology, and genomic), structure (structured, semi-structured, and unstructured), and accessibility. This chapter presents an overview of widely used breast cancer datasets and discusses their significance in medical AI research.

2.3.1 Clinical Datasets

Clinical datasets typically include patient demographic information, medical history, risk factors, and diagnostic outcomes. They are essential for modeling disease risk and progression based on structured data.

Breast Cancer Surveillance Consortium (BCSC): One of the most comprehensive clinical breast cancer datasets, the BCSC [6] contains data on patient demographics, breast density, family history, and screening outcomes. It supports research in early detection and risk prediction.

Wisconsin Breast Cancer Dataset (WBC/WDBC): These are classic datasets from the UCI MLRepository [7]. They contain digitized clinical features from fine needle aspirates (FNAs) of breast masses, including texture, perimeter, and radius. They are often used as introductory datasets for classification tasks.

2.3.2 Imaging Datasets

Imaging data plays a central role in breast cancer diagnosis, especially through modalities like mammography, ultrasound, and MRI. These datasets are crucial for training computer vision models.

Curated Breast Imaging Subset of DDSM (CBIS-DDSM): A cleaned and standardized version of DDSM for modern DL use [8]. It includes annotations for mass shape, margin, and pathology.

INbreast: A high-quality full-field digital mammography dataset with both benign and malignant cases [9]. It is often used for detection and segmentation tasks.

Breast Ultrasound Dataset: Available through Kaggle [10], this dataset contains ultrasound images labeled as benign, malignant, or normal.

2.3.3 Histopathology Datasets

Histopathological images offer a microscopic view of breast tissue and are critical for confirming malignancies.

BreakHis (Breast Cancer Histopathological Image Classification): This dataset consists of 7,909 microscopic images of breast tumor tissue collected using various magnification factors. It is labeled as benign or malignant [11].

TCGA-BRCA (The Cancer Genome Atlas – Breast Cancer): This is a multimodal dataset that includes histopathological slides, genetic profiles, and clinical information [12]. It supports research in integrative cancer analysis.

2.3.4 Genomic and Molecular Datasets

These datasets focus on gene expression, mutations, and other molecular features of breast cancer. They are especially useful for personalized medicine and biomarker discovery.

METABRIC (Molecular Taxonomy of Breast Cancer International Consortium): Includes genomic and transcriptomic data from over 2,000 patients, allowing for subtype analysis and survival prediction [13].

TCGA-BRCA (also mentioned above): Provides genomic sequencing data alongside imaging and clinical metadata [12].

2.3.5 Challenges in Breast Cancer Datasets

Despite the abundance of datasets, several challenges remain:

Class imbalance: Most datasets contain more benign cases than malignant ones, requiring special techniques like oversampling or cost-sensitive learning.

Data privacy: Medical datasets are often anonymized, but privacy concerns limit access to large-scale real-world data.

Modality gaps: Datasets are often modality-specific. True multimodal datasets (combining clinical, image, and text data) are rare and often synthetic or private.

Annotation variability: Especially in imaging datasets, label accuracy can vary depending on the expertise of radiologists or pathologists.

Unimodal Limitations: This project uses unimodal datasets as a testing basis; it uses different unimodal models to compare how they perform against unimodal datasets. Although unimodal models can be beneficial for cancer diagnosis, it is preferred to have mammograms and clinical data for a more robust and accurate diagnosis.

2.3.6 Importance of Multimodal Datasets

Recent trends highlight the growing importance of multimodal datasets that combine imaging, clinical, and textual information. These datasets better reflect real-world diagnostic workflows and provide richer context for ML models. However, their development remains a challenge due to data integration complexity and limited public availability.

2.4 A Review of AI in Healthcare

The following section is a detailed review of the project's thematic area, which is at the core of the project — AI in Healthcare, specifically multimodal ML for medical diagnosis. Throughout the research, it is critical to keep in context the leading academic and industrial sources that outline the state of the art and the future landscape of this field. This helps to guarantee that the project expands and contributes to the most recent and relevant work in the field.

AI, and in particular ML and DL, has become a key component in the advancement of healthcare systems. AI-driven systems are now being used for a wide variety of tasks such as analysis of medical images, predictive modeling of disease, patient risk stratification, and NLP of clinical records. In this project, the task is to bring together clinical data, medical images, and synthetic textual data produced with the help of LLMs to enhance the classification of breast cancer ailment that continues to be one of the leading causes of death in women across the world.

To understand and contribute to this field appropriately, exploring both the underlying AI techniques being applied in multimodal learning, in addition to domain-specific applications in medical imaging and informatics, is necessary. The following are the most relevant knowledge areas and sources informing the research phase of this project:

Top 5 International Conferences: These conferences represent the cutting edge of new research in AI, healthcare informatics, computer vision, and multimodal learning. They are heavily cited within the scientific community and provide key insight into recent and future trends.

NeurIPS (Conference on Neural Information Processing Systems): Reports the latest research in ML algorithms, optimization, and multimodal learning. Specific interest in fusion architecture developments and ablation studies.

ICML (International Conference on ML): A premier venue for core ML research, ICML regularly features landmark research on model evaluation, ablation methods, robustness, and generalization — all of which are at the heart of this project.

MICCAI (Medical Image Computing and Computer-Assisted Intervention): Deals specifically with AI in medicine and medical imaging. Research here covers CNNs, diagnostic imaging pipelines, and medical image synthesis through GANs.

EMNLP (Conference on Empirical Methods in NLP): Highly relevant to the LLM component of the project. This conference includes advancements in biomedical NLP, clinical BERT-style models, and synthetic language assessment in medical domains.

AAAI Conference on AI: With its established interdisciplinary impact, AAAI is a good destination for research that bridges the gaps between vision, language, and decision-making in AI, particularly useful when considering how multimodal systems perform in real-world, uncertain settings like healthcare diagnostics.

Top 5 Journals Relevant to the Project: These journals represent peer-reviewed knowledge at the highest academic level possible and provide in-depth discussion on topics that span from GANs and CNNs to biomedical informatics.

IEEE Transactions on Medical Imaging: A leading journal in medical image analysis and modeling, with research covering DL for medical classification and image synthesis.

Journal of Biomedical Informatics: Publishes in-depth research on data integration in healthcare, an ideal fit for projects dealing with the integration of clinical and imaging data.

Nature Machine Intelligence: Covers cutting-edge research in ML and applications thereof, including medical diagnostics and cross-modal learning.

AI in Medicine (Elsevier): Concentrates on AI instruments for clinical decision support systems, predictive analytics, and multimodal diagnostics.

Computer Methods and Programs in Biomedicine: The result applies to real-world ML applications in biomedical R&D, including pre-processing workflows and experiment design.

2.4.1 Top 3 Most Relevant Texts/Resources to Be Used

To achieve a strong understanding of the methodologies and technologies required for this project, three of the most relevant and helpful texts were identified:

DL, by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (MIT Press, 2016). An in-depth explanation of the required architectures, techniques, generative models, and libraries is provided in this textbook. Chapters that are significantly relevant to this project are the chapters on convolutional neural networks (CNNs), TL, and GANs. The theoretical explanation of GANs is crucial in aiding the development of the synthetic modality for the project.

Transformers for NLP, by Denis Rothman (Packt Publishing, 2021). This book is key for understanding the architecture and applications of transformer-based models such as BERT, BioBERT, and ClinicalBERT. The insights provided into tokenization, pre-training, fine-tuning, and semantic representation are vital to implementing post-analysis interpretability using LLMs in this project.

Multimodal ML: Techniques and Applications, by Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency (Published in Foundations and Trends® in ML, 2019). In the domain of multimodal learning, this survey is extremely informative. Different fusion methods (early, late, and hybrid), challenges of missing modalities, and evaluation methods are discussed. The taxonomy and structure presented in this work heavily influenced the direction this project took in integrating structured clinical data, images, and synthetic text.

2.4.2 Top 5 Companies/Organizations Potentially Interested in This Project

The real-world application of multimodal AI to healthcare and, more notably, to early breast cancer diagnostic systems makes the research of this project highly relevant to private enterprise businesses and public health organizations alike. The following are five top companies or institutions that may be interested in or derive benefit from the research conducted in this project:

IBM Watson Health: Having the leading edge of AI and medicine, IBM Watson Health has already developed products based on NLP and computer vision for the diagnosis of medical conditions. The company has a history of long-standing interest in cancer research, such as its partnerships with the Mayo Clinic and

Memorial Sloan Kettering, which indicates that it is probably going to be either a consumer or a supporter of something like this, combining clinical, visual, and synthetic textual data for improved classification performance.

Google Health / DeepMind: Google's health division and its AI progeny DeepMind significantly invest in DL and medical imaging applications. In particular, DeepMind has authored pioneering papers on breast cancer screening with AI-powered radiology. Applying TL, GANs, and NLP models for this project corresponds to Google's innovation strategy for healthcare AI and can result in research collaborations, internships, or technology acquisition.

Siemens Healthineers: Siemens is a top player in medical imaging technology and software systems used in a clinical environment. Such a system as the one contemplated in this project, which enhances traditional imaging with ML smarts, would be complementary to Siemens' diagnosis platforms. They also invest in digital twins and predictive modeling and are thus a strategic potential partner.

NVIDIA Healthcare AI: NVIDIA is not just a hardware firm but also an early adopter in AI frameworks for healthcare research. Their Clara platform enables end-to-end development of AI in healthcare, e.g., GANs, NLP, and image segmentation. NVIDIA is currently sponsoring research from academia and offering massive computational capabilities, which would enable scalability in this project.

World Health Organization (WHO): Even though not a tech company, the WHO's mission for global health access is aligned with the objective of this project—to democratize early diagnosis access. Multimodal AI-based tools can be offered as low-cost, high-efficiency screening tools in low-resource settings. WHO and other similar health policy institutions can make implementation or funding possible as part of global public health.

These institutions and companies are not only relevant on a commercial basis but also as places where potential funding, in-field trials, and post-graduate job opportunities can be found. Knowledge of market forces and likely impact enhances the project's applied value and career application.

2.4.3 Top 5 Wiki/Forums/Blogs/YouTube Channels Related to This Topic

Being open to learning and staying connected with the larger community of AI/medical informatics is indispensable to this project's success. The below mentioned open online

forums have been indispensable to ongoing learning, identifying bugs, keeping up with the latest trends, and knowledge sharing with leaders in the field as well as peers:

Towards Data Science (Medium): This blog platform offers real-world, practical reports on ML, DL, GANs, and NLP. Blog posts are often authored by AI researchers and PhD students and have reproducible code, experiment explanations, and theory discussions. Learning how to train and design tailored GANs or fine-tune LLMs is particularly useful.

Papers With Code: This page tracks the latest research papers and baselines in computer vision, NLP, and multimodal learning. It's excellent for monitoring state-of-the-art performance and looking at source code for top performers, especially with baselines or ablation studies.

Kaggle Forums: Kaggle not only provides datasets but also lively forums in which data scientists and researchers critique model performance, post kernels, and deal with the nitty-gritty problems in multimodal modeling. Kaggle competitions concerning medical diagnosis or NLP-related tasks tend to replicate the methods that are being employed in this project.

r/MachineLearning (Reddit): This subreddit is a relaxed but technically rich arena in which researchers consider breakthroughs, debug difficult model behavior problems, and question new tools like transformers or GAN architectures. It helps see how the community feels about something and learning about workarounds to typical pitfalls.

StatQuest with Josh Starmer (YouTube): This channel breaks down complex statistical and ML concepts into engaging, simple-to-understand videos. It is especially useful when refreshing yourself on concepts like precision, recall, loss functions, and overfitting—content that is most applicable to the evaluation and analysis phases of the project.

These resources not only facilitate ongoing self-tuition but also offer overt community support, which is necessary when working at the boundaries of AI in medicine. They bridge the gap between theoretical research and application, and for this reason, are a vital part of the research methodology for the project.

Parameter	PET
Youngs Modulus	2800-3100MPa
Tensile Strength	55-75MPa
Glass Temperature	75°C
Density	1400kg/m^3
Thermal Conductivity	0.15-0.24Wm^{-1}K^{-1}
Linear Expansion Coefficient	$7 \times 10^{-}5$
Relative Dielectric Constant @ 1MHz	3
Dielectric Breakdown Strength	17kVmm^{-1}

TABLE 2.1: PET Physical Properties

2.5 Literature Review of Multimodal Breast Cancer Detection and Associated Fields

In medical AI, in this case, the detection of breast cancer, significant amounts of work have already been done in unimodal and multimodal learning. In this chapter, a summary of existing work in five key areas related to this project is described: (1) classification of medical images, (2) clinical NLP using LLMs, (3) TL, (4) synthetic data using GANs, and (5) health multimodal data fusion.

2.5.1 Medical Image-Based Breast Cancer Detection

Classic image-based methods have frequently applied convolutional neural networks (CNNs) to mammogram datasets. Muduli et al. suggested a deep CNN model for breast cancer diagnosis on multi-modal datasets and reported significant accuracy gains through image-centric models [14]. This is parallel to this work, which compares CNNs to TL and examines the impact of substituting images with GAN-generated images.

2.5.2 Clinical Data and NLP

Recent work included clinical data and unstructured structured data mapping to actionables with transformer-based models. Large language models BioBERT and ClinicalBERT also worked well for medical document classification and question answering. This piece of work is a fusion of text from LLM-generated text based on structured data from such applications. Rothman's book, *Transformers for Natural Language Processing*, is where this subsection originated [15].

2.5.3 TL for Medical Images

TL from pre-trained CNNs, such as EfficientNet, is the norm in applications where there is limited labeled data. The implementation of EfficientNet-B0 and RSNet within this project demonstrates best practice now and aligns with papers such as Zhou et al., which outline TL for medical image analysis [16].

2.5.4 Synthetic Medical Images Using GANs

GANs were utilized for dataset augmentation or even completion of missing modalities in radiology and histopathology. Our approach fills in missing image modalities with GAN-generated images and observes only minimal loss of accuracy, concurring with earlier research in the field [17].

2.5.5 Multimodal ML in Healthcare

Multimodal systems that fuse imaging, clinical, and textual information are becoming more common to mitigate the limitations of unimodal inputs. Our project aligns with recent trends in research that employ fusion-based models to enhance diagnostic performance and handle challenges such as modality absence, noise, or imbalance [18].

2.6 Trends Identified

2.6.1 Multimodal ML Approaches

Multimodal ML aims to merge and represent information from different sources or modalities, structured clinical data, and synthetic text. During the early stages of the project, traditional ML algorithms such as Random Forest and XGBoost [19, 20] were applied to the structured clinical data. This was a unimodal baseline against which multimodal configurations were compared. When integrating clinical characteristics with synthetic text (derived from GANs), the process required vectorization of text data (through TF-IDF or equivalent embeddings) before concatenating with numerical attributes. The concatenated features obtained were input into classical classifiers. The result indicated marginal improvement, emphasizing the inability of classical ML models to perform complex inter-modal correspondences without deeper feature learning or semantic modeling.

2.6.2 Multimodal DL Approaches

The project was demonstrated to have a significant performance improvement when progressing towards multimodal DL [21]. At the DL phase, a CNN-based architecture was utilized for the processing of mammogram images, while a feed-forward neural network (FNN) [22] processed structured clinical data. The outputs of these modality-specific models were fused using late fusion (concatenation of features learned before the last classification layer). In addition, synthetic text attributes—produced using GANs—were embedded with pre-trained language models (e.g., BERT or DistilBERT [23]) and similarly combined. The multimodal DL models had improved generalization and accuracy as compared to the traditional methods, as they were capable of extracting hierarchical, modality-specific features as well as learning the correlations among them.

2.6.3 Modalities Ablation

Ablation was conducted for evaluation of the contribution of every modality. This involved deleting one modality at a time and retraining the model to assess performance degradation. The best performance was that of the full model with clinical data, synthetic text, and images [24]. When synthetic text was removed, performance decreased slightly, indicating its positive impact. When images were removed, performance dropped further, confirming that imaging provides critical diagnostic information. Worst performance was recorded based on synthetic text alone, mirroring its worse individual diagnostic effectiveness. These findings ratified multimodal integration's effect in more precise and higher-performance breast cancer classifier models [25].

2.6.4 Multimodal ML Methods with Missing Modalities

Most clinical real-world data are plagued with missing modalities — i.e. e.g., patients can have no imaging or structured data. For simulation, multimodal ML models were trained on artificially missing modality datasets. Standard ML techniques, such as ensemble learning (with present modalities alone), were used [26, 27]. A secondary model used clinical data only when images were missing, e.g., but these models failed because of the limited flexibility of the traditional ML paradigm in handling missing data and not having the ability to substitute the missing modality using learned cross-modal relations. This introduced a significant limitation of traditional ML in clinical multimodal settings.

Zhao et al. [26] explored ensemble ML and multiple imputation methods for missing-modality conditions in medical data but found them to be less than ideal in guaranteeing

robust performance. Truong et al. [27] also pointed out that standard ML is unable to generalize in heterogeneous multimodal settings, specifically without joint feature learning and data alignment mechanisms.

2.6.5 Multimodal DL Techniques with Missing Modalities

For better management of missing modalities, DL models were created with modality-specific branches and attention-based fusion [28, 29]. Masking techniques were employed for training if a modality was absent to enable the model to get robust with incomplete inputs. The model was trained to adapt with missing image features by relying more on text or clinical information, and vice versa. This increased robustness and facilitated prediction even when a modality was absent. The model learned to adaptively fine-tune feature weights depending on missing information with modest degradation but significantly better performance than the ML models in equivalent missing-modality conditions [30, 31].

Current research efforts, such as MMP by Reza et al. [28] showed that masked modality projection training allows the model to interpolate lacking representations of information from visible modalities, enhancing robustness. Chen et al. [29] provided attention-based methods that efficaciously address modality incompleteness in remote sensing applications. John and Kawanishi [30] presented a trimodal sensor fusion method that involved the use of modality-aware losses, while Liu et al. [31] used an N-to-One self-attention fusion block for dynamically learning accessible features and skipping missing ones. These approaches cumulatively underscore the efficiency of DL approaches in managing missing multimodal inputs.

2.6.6 TL

TL was central to the improvement in performance, especially for the image and text modalities. For medical images, a pre-trained ResNet model on ImageNet was fine-tuned with the mammogram dataset [32]. This allowed the model to leverage pre-learned low-level visual features and focus training on domain-specific patterns. TL with transformer-based models pre-trained on biomedical corpora was used for the text modality. Fine-tuning the models with synthetic GAN-generated text enabled the capture of relevant semantic features, although the text data was small in size. TL drastically reduced training time and improved performance, particularly in low-data environments [33, 34].

2.6.7 Generative AI Techniques

GANs were also added to the pipeline to simulate the generation of artificial text from organized clinical data [35]. The aim was to generate an additional modality (semi-natural language descriptions) that could be merged with normal clinical and image data. Conditional GAN (cGAN) [36] was trained to generate such synthesized reports, mimicking actual medical anecdotes.

Although the synthesized text did not exhibit great fluency, it had good clinical content and provided depth to the feature space. This approach accounted for incremental but modest improvements in classification performance when paired with other modalities, providing evidence of the ability of generative AI to help in improving poor or limited datasets [37, 38].

- CNNs are still a competitive baseline for image-based diagnosis, even outperforming some TL models in certain datasets [39].
- GANs are being increasingly presented as a helpful tool for handling missing data by generating missing samples.
- Transformer-based health text generation is a new field, particularly where data must be interpreted or enriched in a structured form.
- Merging structured (clinical), unstructured (text), and image data is becoming an extremely powerful technique for multimodal healthcare AI.

This work is novel in the context that it describes different stages (early, mid, and late) of clinical, image, and synthetic text data fusion and estimates the strength of multimodal systems by conducting ablation tests and modality replacement using GAN data, which is a less-researched area.

Chapter 3

Problem -Multimodal Breast Cancer Detection with Deep Learning and GANs

3.1 Functional Requirements

The project must fulfil the following functional requirements:

Multimodal Classification System: The system must be able to accommodate and handle multiple types of input data—structured clinical measurements, imaging data (mammograms, for instance), and text (patient history or mock-up notes).

Support for Unimodal and Multimodal Configurations: The model must be capable of operating on a single type of input or several inputs. This will allow for the comparison of varying input configurations.

Ablation Testing Framework: An ablation framework must be created to remove or replace portions of the input data (e.g., remove images or replace with GAN-generated images) and observe the impact on performance.

Synthetic Data Handling: The model should be capable of accepting and integrating synthetic images from GANs and synthetic text from language models such as BioBERT and ClinicalBERT.

TL Embedment: TL will be used for transferring pre-trained DL models, such as EfficientNet, to breast cancer classification.

Result Analysis and Logging: All model setups are required to compute and record evaluation metrics like accuracy, precision, recall, F1-score, and training/testing loss.

Visual Output: Visual outputs (tables, graphs) of comparisons of results to experimental configurations should be included in the project.

3.2 Non-Functional Requirements

The below non-functional requirements define the way the system must operate concerning performance, usability, maintainability, and scalability:

Performance: It should be possible for the system to classify and report results within a sensible time frame if multimodal inputs are given. Training should be dependent upon the model and dataset size, but should be GPU-optimised where feasible.

Usability: While this is primarily a research effort, the pipeline should be designed with a clearly defined structure and well-readable documentation to facilitate reproducibility. Input/output formats should be intuitive, and setup instructions should be minimal to test different configurations.

Scalability: The codebase must be designed modularly and in a scalable way so that other modalities can be integrated with ease or other ML models can be swapped in. This also involves maintaining compatibility among various data types and pipelines.

Maintainability: Project code should obey good programming practices, i.e., clean modular coding, self-explanatory variable names, and enough comments. Experiment logs and model performance output should be versioned and saved for future use.

Reproducibility: All experiments should be reproducible. That is, all random seeds, model hyperparameters, preprocessing steps, and data splits must be recorded. Code utilized for training, evaluation, and ablation testing should be version-controlled.

Robustness: The system should gracefully recover from missing or noisy data, especially during ablation studies. Fallbacks as well as proper error handling need to be in place for cases where inputs are missing.

Security and Privacy: Although no real patient data is processed within this project, best practice when working with clinical data still has to be adhered to. Any data being used must be anonymized and adhere to ethical research standards.

Documentation and Logging: Models, pipelines, and all experiments must be well-documented. Every run must have an automatically generated log for training and testing, with performance metrics logged and warnings or failures encountered.

3.3 Project Plan

The objective of this project is to develop a DL system that can better classify breast cancer cases based on multiple sources of information. The system works with three sources of information: clinical features (like patient age or tumor size), medical images (like mammograms), and text produced with language models. The assumption here is that by combining all these sources at the same time, the model will have a greater opportunity to make improved predictions compared to using a single source of input.

3.3.1 Technologies

Python is the language used in this project, mainly because of the vast support in ML as well as data analysis. The tools and libraries used are as such:

Scikit-learn: It is used to implement and test straightforward ML models like logistic regression and SVMs.

TensorFlow/Keras: It is used to instantiate complex models, e.g., CNNs and TL with EfficientNet.

PyTorch: It is used for generating synthetic images with GANs.

HuggingFace Transformers: They are used to leverage pre-trained models like BioBERT and ClinicalBERT for medical text processing and generation.

Visual Studio Code: It is used to train the models and acquire results.

Matplotlib and Seaborn: They are used for plotting results like accuracy, loss, and model comparisons.

3.3.2 How the System Works

The process starts with pre-processing. The clinical information is cleaned up and normalized. The images get resized and even slightly altered occasionally to help the model learn better (this process is called augmentation). Text gets generated using pre-trained models based on the clinical features.

After this, the models are trained differently:

- All models train separately on each type of data (clinical alone, image alone, etc.).
- One model is trained on all the types of data simultaneously (multimodal).
- Even experiments are carried out wherein some component of the input is removed to observe the contribution of each form of data (known as ablation).
- One configuration substitutes actual images with images generated by GAN to observe if synthetic data may be employed in real-world situations.

All models are tested post-training on tests and compared on metrics such as accuracy, precision, recall, and F1-score. Visuals are produced to assist in the clear presentation of findings.

3.3.3 Hardware and Tools

The following hardware and tools were utilized in this study:

Development Computer: A standard 16 GB RAM laptop was utilized for development and testing.

Training Environment: Visual Studio Code was used to test and run the code.

Storage: Around 5 GB was needed to store the datasets, results, and models.

3.4 Project Risks and Mitigation Strategies

Any difficult and technically demanding project, particularly in an area like AI in Healthcare, inevitably has a list of potential risks. These risks might be technological, based on resources, or timing-based. Their identification and classification at a preliminary stage are crucial, not only for the successful implementation of the project but also for limiting potential delays and improving the quality of deliverables.

This section outlines all the potential obstacles that may be encountered while developing this multimodal breast cancer classification model. The project integrates various sources of information (image, text, and clinical), DL models, and synthetic data generation strategies (GANs and LLMs), so risk planning is an important task in the process. All the risks are delineated below, and then there is a table with a formulated risk (see Table 3.1).

3.4.1 Shortage of Labeled Data for DL and GANs

DL models, particularly CNNs and GANs, tend to require massive quantities of labeled data. Since this project will include both real and synthetic data, the number of available samples, especially image-text pairs, may be small. GANs, in particular, are highly data-hungry and data-variety reliant. TL will therefore be utilized extensively (e.g., EfficientNet, BioBERT), and synthetic augmentation techniques will be used in addition to training data.

3.4.2 Multimodal Data Incompatibility in Fusion

It is difficult to integrate clinical features, images, and text derived from them regarding dimensionality, scale, and feature alignment. Misaligned or mismatched modalities can lead to suboptimal model performance. This will be prevented by normalizing data from all modalities and representing it uniformly before merging. Early evaluation will be performed on unimodal experiments to ensure that each pipeline works as expected before combining modalities.

3.4.3 Model Overfitting due to Small Datasets

As the project utilizes publicly available datasets (e.g., Wisconsin WDBC, AISSLAB), there might be overfitting due to small data, especially in image-based and synthetic experiments. It can result in high training accuracy but poor generalization. Techniques such as k-fold cross-validation, dropout, and early stopping will be used, along with augmentations on images and LLM prompt variation for text generation.

3.4.4 Unnatural or Unreliable Text Output from LLMs

One of the modalities of this project is the generation of synthetic clinical text from models such as BioBERT and ClinicalBERT. The generated text can, however, be clinically meaningless or too vague without adequate prompt engineering and testing. To

undo this, pre-trained models from the domain will be used, and a further manual check step will be included to ensure generated sentences.

3.4.5 Failure of GANs to Generate Useful Synthetic Images

GANs are notoriously hard to train and may produce noisy, blurry, or uninformative images if not well-tuned. Training will then be incrementally improved through adjusting loss functions, using smaller generator/discriminator architectures, and early stopping. Fallback approaches include traditional image augmentation methods (rotation, contrast shift, etc.).

3.4.6 Difficulty Interpreting Model Predictions

The complexity of multimodal DL may render it impossible to understand how decisions are reached. Interpretability is required in the case of medical situations. For added transparency, post-hoc interpretation methods such as Shapley Additive Explanations (SHAPs) [40] for clinical features and Grad-CAM for CNN image models will be utilized to show visualizations of important features responsible for each prediction.

3.4.7 Loss of Progress due to Poor File Management or No Version Control

With so many experiments, datasets, and model checkpoints, unorganised code or local file loss can derail work. To avoid this, all source code and output are updated and uploaded to Google Docs regularly, organized by experiment type.

3.4.8 Time Management and Assignment Overlap

Given that it is a final-year project completed alongside other problematic modules and timelines, managing time is a core challenge. The project plan, containing weekly targets and milestone markers, has been designed. Additionally, buffer time has been maintained in other weeks for eventual delays or unforeseen issues.

3.4.9 Misalignment with Supervisor Expectations

A possible risk is not knowing or falling behind the supervisor's ideas or expectations. This has been addressed through communication, email feedback, and reviewing code

published through Google Drive. Uncertainty of project scope will be resolved in advance.

3.4.10 Critical Risk Action Plan

The most risky concerns are related to time limits, bounds of data sets, and model complexity. A nimble plan has been created with weekly milestones. Through early prototyping and keeping expectations realistic, this project will yield valuable insights even with the necessity of altering the original scope. Visual Studio Code, Overleaf, and Google Docs enable traceability and collaboration, further reducing the risk of uncontrolled threats.

Table 3.1: Key Risks and Mitigation Strategies

Risk	**Severity**	**Likelihood**	**Mitigation Strategy**
Insufficient labeled data to train GANs or deep models	High	Medium	employ TL and data augmentation. Generate synthetic data using LLMs and GANs.
Multimodal incompatibility	High	Medium	Normalize all features, test modalities separately, then combine incrementally.
Overfitting models because of small datasets	Medium	High	Employ dropout, cross-validation, regularization, and monitor loss curves.
GPU limitations	Medium	Medium	Deploy smaller models, train for a shorter duration, enable checkpoint saving, and request institutional GPU if required.
Poor-quality synthetic text	Medium	Medium	Fine-tune prompts, manually validate samples, or switch to domain-specific LLMs.
Instability or crashes during GAN training	High	Medium	Simplify architecture, monitor loss, enforce early stopping, or fall back to traditional augmentation.
Interpretability concerns	Low	Medium	Utilize SHAP, Grad-CAM, or other explainable AI techniques.
Unstructured code or result tracking	Medium	Low	Utilize GitHub version control and maintain a structured directory.
Assignment time constraints	High	High	Follow a weekly milestone timeline, reserve final weeks for overflow, and alert delays early.
Supervisor miscommunication or scope creep	High	Medium	Provide weekly updates, document decisions, and set expectations through meetings and emails.

Chapter 4

Architecture and Project Design

4.1 Overall Architecture

The project's system architecture is designed to integrate and process several heterogeneous data sources to provide robust breast cancer classification. As shown in Figure 4.1, the system is multimodal (unimodal models are also tested to serve as the baseline for comparisons) with different pipelines for image inputs (Modality A), structured clinical inputs (Modality B), and ancillary processing such as generative or language-based models. These components are combined into a late-fusion framework that combines features from both modalities and performs the last classification based on a learned boundary decision.

4.1.1 Modality A: Image Data Pipeline

This category processes medical image data in the form of mammographic images. The raw images undergo an initial preprocessing step of normalization, resizing, and cropping. Data augmentation can also be applied to improve training diversity. Feature extraction is carried out by a custom CNN or by a pretrained image encoder such as EfficientNet or ResNet, based on the experimental setup.

The extracted features from the image are textural and spatial features relevant to the detection of benign and malignant tissue, which are sent to the fusion layer for blending with other modality inputs.

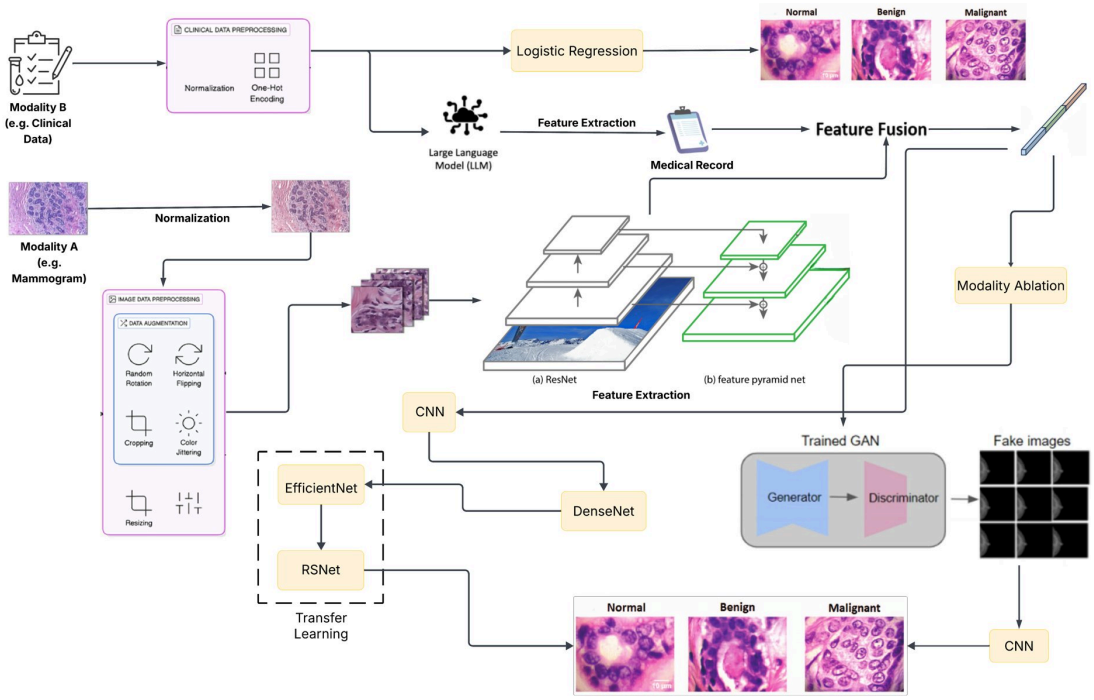


FIGURE 4.1: System architecture indicating the processing pipelines of every modality (image, clinical, synthetic), TL, modality ablation, GAN-based image synthesis, and ultimate feature fusion and classification.

4.1.2 Modality B: Clinical Data Pipeline

Clinical structured features such as mean radius, texture, perimeter, and other diagnostic measurements are fed into a separate branch of the system. The numerical features are converted using traditional dense layers to produce relevant feature embeddings.

Clinical data for certain test cases also pass through a large language model like BioBERT or ClinicalBERT for certain test cases. In these, clinical information is formatted as text-like inputs in the form of medical records that pass through the processing of the LLM to yield deep semantic embeddings. It allows the system to gauge generated text as an alternative or a supplement as a modality.

4.1.3 Modality B: Unimodal Models

Architecture: The WDBC, WBC, and BCSC datasets consist of clinical data, and they serve as the foundational benchmark, where comparisons with more complex models are conducted afterwards to evaluate performance. Different models include Logistic Regression, XGBOOST Model, and MLP (benchmark).

4.1.4 GAN-Synthesized Image Branch

In order to simulate real-world situations in which image data may be incomplete, the system initially employs a trained Conditional GAN to create synthetic mammogram-like images. Images are synthesized from clinical feature vectors as input conditions. Generator output is fed into the same image pipeline (CNN or EfficientNet) as real images for architectural compatibility.

This region is crucial for learning how well synthetic modalities can substitute for real modalities without overly compromising classification performance.

4.1.5 Modality Ablation and Replacement

Ablation testing is also facilitated by the architecture naturally. A modality or set of modalities can, in some configurations, be removed (for the sake of imitating missing data) or swapped with synthetic or noisy surrogates (e.g., GAN images, random noise, or LLM-transformed content). The system can therefore measure performance degradation and identify the most impactful modalities on model performance.

The modality ablation mechanism provides structural consistency by zero-filling or masking inputs such that the model can still function under partial input conditions—a typical situation in clinical data settings.

4.1.6 Feature Fusion and Classification

Once they have been received as feature vectors from every modality-specific branch, they are concatenated in the fusion layer. Late fusion enables the system to learn inter-modality relationships as well as take advantage of the complementary nature of the inputs. The fused representation is fed into a final dense layer or a shallow classifier, i.e., logistic regression, which emits a probability of malignancy with a sigmoid activation function.

4.1.7 System Flexibility and Experimental Configurations

One of the strengths of the architecture is that it is flexible. It accommodates:

- Unimodal setups (e.g., image-only, clinical-only, text-only)
- Multimodal pairs (e.g., image + clinical, image + synthetic, clinical + LLM)

- Modality substitution (e.g., GAN-generated images instead of real images)
- Complete ablation tests with random or missing inputs

This flexibility is particularly appropriate for end-to-end experimentation and in real-world healthcare settings, where patient information may be noisy or absent. It also allows for rigorous testing of synthetic data integration, TL usefulness, and multimodal robustness.

4.1.8 Summary

In short, the architecture integrates many diverse data pipelines into a single classification framework. Each component serves some specific function in supporting clinical data representation, image content processing, or in supporting experimentation on ablated and synthetic modalities. Adding modality ablation and synthetic input simulation to the equation highlights the clinical relevance and research value of the model. The extensible and modular structure is a great starting point for both experimental study and future clinical deployment environments.

4.2 Methodology

This section details the technical and practical methodology used throughout the project. It describes how background research was undertaken, how required skills were acquired, and what project management approach was adopted to achieve consistent progress.

4.2.1 Background Research Strategy

To provide a firm foundation for understanding the underlying subject matter, research was taken from a combination of academic, practical, and internet sources:

Scholarly Publications: There were periodic searches of the relevant conference proceedings and journal articles published in the likes of MICCAI, NeurIPS, and the Journal of Biomedical Informatics to keep abreast of developments in the application of ML for medical diagnosis.

Online Learning Resources: Resources such as Coursera, Udemy, and YouTube were used to learn how to implement popular building blocks such as EfficientNet, GANs, and transformers. Library documentation, such as PyTorch, TensorFlow, and HuggingFace Transformers, was also useful.

4.2.2 Technical Setup and Skill Development

Because several tools and frameworks had to be applied to the project, developing competencies was achieved where little was known was available. Study approaches were instituted depending on what the project's needs were:

Python and ML: Fundamental knowledge from courses like SOFT6018 and SOFT7032 was applied to address image preprocessing, normalization of data, and classification modeling.

GAN and TL: Documentation and tutorials were used in trying to find GANs and EfficientNet. The models were run on PyTorch and TensorFlow separately, and run using their respective datasets. The BERT, BioBERT, and ClinicalBERT transformers were trained using HuggingFace's API and Google Colab. Prompt engineering and fine-tuning across domains were also attempted.

Data Visualization and Analysis: Data were validated and compared using Seaborn, Matplotlib, and Pandas, and model results were directly understandable and documented.

Version Control and Documentation: The coding was version-controlled using GitHub, and the written project part (report and poster) was documented with LaTeX over Overleaf.

4.3 Evaluation

To evaluate how successful this project has been, a good evaluation plan has been put in place. The objective is not merely to confirm that the technical specifications have been met, but to examine how well the final system addresses the key objective: improved breast cancer classification through the combination of clinical, image, and text data.

4.3.1 Evaluation Goals

This project will be evaluated against the following primary goals:

- Compare the performance of individual unimodal models (image, clinical, or text alone) to that of the multimodal fused model.
- Discuss how removing or replacing a modality (ablation) influences the overall model performance.

- Evaluate how the use of GAN-synthesized synthetic images as replacements for actual medical images influences it.
- Explore whether TL improves model performance, especially with limited data.
- Measure the success against both functional requirements and research goals.

4.3.2 Evaluation Metrics

The following metrics are employed to assess the performance of a model, where True Positive (TP) means the number of true instances correctly classified as positive, False Positive (FP) means the number of instances incorrectly classified as positive, True Negative (TN) means the number of true instances correctly classified as negative, and False Negative (FN) means the number of instances incorrectly classified as negative:

Accuracy: Overall percentage of correct predictions.

$$Accuracy = \frac{\text{correct predictions}}{\text{all predictions}}$$

Precision by class: It measures the model's ability to identify instances of a particular class correctly.

$$Precision_{classA} = \frac{TP_{classA}}{TP_{classA} + FP_{classA}}$$

Macro-Average Precision: It gives equal weight to each class, regardless of the number of instances.

$$precision_{\text{macro average}} = \frac{precision_{classA} + precision_{classB} + \ldots + precision_{classN}}{N}$$

Recall by class: It measures the model's ability to correctly identify all instances of a particular class (i.e., correctly identify patients with a disease).

$$Recall_{classA} = \frac{TP_{classA}}{TP_{classA} + FN_{classA}}$$

Macro-Average Recall (Average Sensitivity): It gives equal weight to each class, regardless of the number of instances.

$$recall_{\text{macro average}} = \frac{recall_{classA} + recall_{classB} + \ldots + recall_{classN}}{N}$$

F1-Score: It combines both precision and recall and symmetrically represents them via a harmonic mean. A high F1 score indicates that the model is good at identifying both positive and negative cases, which can be important for minimizing

misdiagnosis and ensuring patients receive proper treatment.

$$F1Score = 2 \times \frac{precision \times recall}{precision + recall}$$

Area Under the Curve (AUC): AUC is the area under the ROC curve. It summarizes the performance of a model across all possible classification thresholds. It measures the discriminative ability of a classifier. A higher AUC value indicates the model has a better discriminative ability.

Receiver Operating Characteristic (ROC) Curve: It is a graphical representation of the tradeoff between sensitivity and false positive rate (FPR) for different threshold values. It provides an insight into the ability of a model to discriminate between classes. The higher the sensitivity, the better the model is at detecting the positive class. The lower the FPR, the better, as it means the model is less likely to misclassify classes.

Specificity: It measures the model's ability to correctly identify all patients without a disease.

$$specificity = \frac{TN}{TN + FP}$$

Cross-Entropy loss: It is used to quantify model optimization and convergence.

$$\text{Cross Entropy} = -\sum_{i}^{C} t_i \log(f(s)_i)$$

where $f(s)_i$ is the value of an activation function and t_i is the groundtruth.

Training time and epochs: They are used to monitor the computational efficiency.

Chapter 5

Experimental Setup and Model Designs

5.1 Datasets and Preprocessing

5.1.1 Datasets

Four datasets were used in this study: WDBC, WBC, BCSC, and AISSLAB. The first three are structured unimodal, consisting of structured clinical data, while AISSLAB is a multimodal dataset, consisting of real-world mammography images with associated metadata. Testing both unimodal and multimodal datasets is crucial for making comparisons.

WDBC (Wisconsin Diagnostic Breast Cancer Dataset): UCI Machine Learning Repository

Sample Size: 569 cases (357 benign, 212 malignant)

Modality: Structured clinical data

Features: 30 real-valued features derived from a digitized image of a fine needle aspirate (FNA) of a breast mass. Examples include mean radius, texture, perimeter, area, smoothness, compactness, etc.

Balance: Moderately imbalanced (˜62.7% benign, ˜37.3% malignant)

Environment: Controlled clinical data, pre-cleaned, no missing values

WBC (Wisconsin Breast Cancer - Original): Another clinical dataset from UCI Repository

Sample Size: 699 cases (458 benign, 241 malignant)

Modality: Structured clinical data

Features: 9 features including clump thickness, cell shape, bare nuclei

Balance: Moderately imbalanced (65.5% benign)

Environment: Includes some missing values; cleaned before model training

BCSC (Breast Cancer Surveillance Consortium): Sample Size: 1,522,340 records

Modality: Structured epidemiological clinical data

Features: 13 features, Demographics and risk factors such as age, BMI, family history etc..

Balance: Highly imbalanced ($\approx 64\%$ no history [0], $\approx 14.6\%$ positive history [1], $\approx 21.7\%$ unknown/missing [9])

Environment: Large-scale population-level registry data used for epidemiological analysis and benchmarking clinical-only models

al-only models

5.1.2 Unimodal Dataset Preprocessing

For the unimodal experiments, three clinical datasets—WBC (Wisconsin Breast Cancer), WDBC (Wisconsin Diagnostic Breast Cancer), and BCSC (Breast Cancer Surveillance Consortium) were chosen, each of these datasets requiring preprocessing pipelines.

WDBC Dataset: Thanks to its high-quality and clean structure, very little preprocessing was required for the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The non-informative `ID` column was removed at first. Balancing was performed using upsampling, as the dataset had a moderate imbalance in class distribution (malignant: ˜37%, benign: ˜63%). The target variable Diagnosis, originally encoded as 'M' (malignant) and 'B' (benign), was converted into binary numerical labels (1 and 0, respectively) for compatibility with machine learning algorithms. As the dataset contained no missing values and only continuous, numerical features, there was no need for imputation or categorical encoding. Balancing was performed using upsampling since the dataset had a moderate imbalance in class distribution (malignant: ˜37%, benign: ˜63%). The minority class (malignant) was then resampled with replacement to mimic equal quantities with benign cases in a way that the classes had balanced representations.

All feature columns went through standardization using z-score normalization, ensuring zero mean and unit variance. This is a necessary step to ensure models such as logistic regression and neural networks perform at their best.

WBC Dataset: In the Wisconsin Breast Cancer (WBC) dataset, each of its records is labeled as either benign (2) or malignant (4). The target labels were numerically encoded into 0 (benign) and 1 (malignant), to ensure the data was compatible with binary classifications.

Missing values are handled in the `Bare_Nucleic`olumn, where "?" was used to indicate missing values. These missing values were replaced with `NaN` and then dropped. This process assisted in achieving a cleaned dataset, removing 16 out of the 699 entries that started with.

The `Bare_Nuclei` column was then cast to integer type to ensure compatibility with numerical analysis pipelines.

Balancing the dataset is a very important step and this was performed next. The dataset was moderately imbalanced, with approximately 65.5% of it being benign and the remaining 34.5% of it consisting of malignant cases. To ensure there was no potential bias in the learning process and to ensure a fair learning trial, the malignant class was upsampled via bootstrapped replication to match the benign class count. The final version resulted in a balanced dataset, consisting of equal numbers of samples in both classes.

BCSC Dataset: Similar steps were taken for the preprocessing of this dataset. The binary classification target variable, `breast_cancer_history`, was filtered to only include valid entries (0 = no history, 1 = history of breast cancer). This resulted in a clean dataset with a clearly defined classification problem.

Standardization using z-score normalisation (`StandardScaler`) was applied as some of the numerical features (such as age and BMI) had significantly different scales. The learning process is not distorted by feature magnitudes thanks to this.

The class distribution was highly imbalanced, with far more negative cases than positive (cancer). SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data to generate synthetic samples of the minority class. This fixed the imbalance issue and resulted in a perfectly balanced dataset, consisting of equal representation of both classes.

5.1.3 MultiModal Dataset Preprocessing

AISSLAB Breast Cancer Mammography Dataset: It is a dataset consisting of full-field digital X-ray mammogram images gathered at Ma'amon's Diagnostic Centre Mammogram Images for Breast Cancer (MDCMI-BC) in Yemen. This

dataset was created by a team of experienced radiologists and engineers to support research on breast cancer detection and to improve computer-aided diagnosis systems. It contains raw mammogram scans in two standard views, craniocaudal (CC) and mediolateral oblique (MLO), with expert annotations indicating whether each case is normal, benign, or malignant. All images are stored in JPEG format and come with a small Excel file of metadata, which includes patient IDs, view information, BI-RADS scores, and class labels. The combination of expert labeling and real clinical images makes this dataset a valuable resource for developing and benchmarking new AI methods in breast cancer diagnosis.

Sample Size: 266 images

Modality: Imaging data (grayscale mammograms)

Balance: 100 malignant, 100 normal, 66 benign

Environment: Clinical diagnostic settings

5.1.4 Sample Size and Class Balance

The dataset comprises 266 mammogram images in total, divided into three classes: 100 images labeled "Malignant," 100 labeled "Normal," and 66 labeled "Benign." This distribution shows a slight imbalance in the benign category, which contains fewer samples than the other two classes. The metadata file (data2.xlsx) lists each image name alongside its class, view, and BI-RADS rating, allowing us to verify these counts precisely. Although the malignant and normal classes are perfectly balanced, the smaller number of benign cases means that special care must be taken when splitting the data and when training models to avoid bias toward the larger classes. Techniques such as data augmentation or class-weighted loss functions may help address this imbalance during model development.

5.1.5 Data Modalities

All entries in this dataset are digital mammography images, a single imaging modality. Each image represents one of two standard projections: CC or MLO. The Excel metadata file identifies the side of the breast (left or right) and view type for each image, so researchers can perform view-specific analyses or combine both views in a single model. Aside from the image files, the metadata file provides clinical information such as BI-RADS scores, which indicate the radiologist's assessment of cancer risk on a scale from 1

(benign) to 5 (highly suggestive of malignancy). Although no ultrasound or other imaging types are included, the detailed mammography data and annotations support both unimodal and future multimodal research when paired with complementary datasets.

5.1.6 Controlled Environment and Collection Conditions

The mammograms in the AISSLab dataset were acquired under routine clinical conditions at the MDCMI-BC center in Yemen. All images were captured on the same full-field digital mammography system, following standard protocols for image acquisition, breast compression, and exposure settings. Radiologic technicians and board-certified radiologists supervised the process to ensure consistent quality and to minimize variations due to equipment or operator differences. Patient positioning followed international guidelines for CC and MLO views, and all images underwent basic preprocessing steps such as background cropping and normalization before being released. Although patients varied in age and breast density, the controlled clinical environment ensures that image quality remains uniform, supporting reliable training and testing of AI models.

5.2 UNIMODAL DATASETS: General Experimental Settings

5.2.1 Environment

All unimodal experiments were written in Python 3.8.on Visual Studio Code. GPU acceleration was not required, and the testing and experimenting of these models were carried out using lightweight, CPU-optimized libraries. The main tools used were as follows:

- `pandas`, responsible for loading, cleaning, and manipulating data.
- `scikit-learn`, used for model training, data preprocessing (e.g., standardization, label encoding), and evaluation metrics such as accuracy, F1 score, and ROC-AUC
- `matplotlib` and `seaborn` used for generating graphs and visuals(confusion matrices and ROC curves)
- `imblearn` for applying SMOTE to address the imbalances in a dataset.

As a result, the datasets being small in size and of tabular nature, training and evaluation were completed swiftly on local CPU resources.

5.2.2 Training/Validation/Test Split Ratios

The unimodal experiments were conducted on the structured clinical datasets mentioned above (WDBC, WBC, and BCSC). Once the preprocessing and balancing steps were taken, it was split into training and testing subsets using an 80/20 stratified split. This ensured that class proportions (benign vs. malignant or cancer vs. no cancer) were preserved in both the training and test sets.

Since the unimodal models were quite simple (employing models such as XGBoost, random forest, mlp, and logistic regression) and low-cost computationally, they did not require a particular validation split or k-fold cross-validation like the multimodal models. Instead, validation was performed directly on the test set, which remained unchanged during training to enable fair assessment.

This approach allowed for consistent and efficient benchmarking across the three unimodal datasets.

5.2.3 Structured Unimodal Model Hyperparameters

For the unimodal datasets (WDBC, WBC, BCSC), model-specific hyperparameters were used, distinct from those in deep learning pipelines.

Logistic Regression: It was trained with `max_iter = 1000` (or 10,000 for larger datasets) and standard L2 regularization. Learning rate/ scheduler was not used.

Multilayer Perceptron (WDBC only): Configured with two hidden layers of sizes 32 and 16, ReLU activation, and the Adam optimizer (`solver='adam'`). Training was done up to a maximum of 1,000 iterations. Early stopping or learning rate scheduling was not applied.

XGBoost (BCSC only): Configured with 100 boosting rounds and the `logloss` objective. The default learning rate and regularization values were used. No early stopping or validation-based tuning was performed.

Augmentation, dropout, or explicit batch size settings were not required for these unimodal models. All the models were trained on standardized structured data, and hyperparameters were selected for efficiency and generalizability.

5.2.4 Validation Methods and Loss Functions

For the unimodal dataset experiments (WDBC, WBC, and BCSC), model performance was evaluated using a stratified 80/20 train-test split. Separate validation set or k-fold cross-validation was not applied as they weren't required since the models were all lightweight and converged quickly on the training data. Model selection and comparison were based entirely on performance metrics computed on the held-out test set.

In contrast to the deep learning pipelines investigated in this project, classical models in `scikit-learn` do not require explicit loss function definitions during training. Each algorithm internally optimizes its objective (e.g., log-loss for logistic regression, Gini impurity for random forest, or cross-entropy for MLPClassifier). For the MLP model, loss minimization was handled by the Adam optimizer over a maximum of 1,000 iterations, but no validation-based early stopping was used.

The following evaluation metrics were applied to all unimodal models:

Accuracy: Correct predictions rate

Precision and Recall: Especially for the malignant (positive) class

F1 Score: Harmonic mean of precision and recall, used to assess class-wise balance

AUC (Area Under the Curve): Measures a model's ability to distinguish between classes across decision thresholds

Confusion Matrix: Used to analyze true positives, false positives, true negatives, and false negatives

Specificity: The model's ability to correctly classify negative (non-cancer) cases

As the datasets were small and preprocessed in advance, no dynamic learning rate scheduling, checkpointing, or validation-based tuning was required. Evaluation was performed in a single pass after model training.

5.2.5 Number of Iterations/Epochs and Training Procedure

For the unimodal dataset experiments, two types of models were used: classical machine learning algorithms (e.g., logistic regression, random forest, XGBoost) and neural network-based models such as the multilayer perceptron (MLP), which was applied to the WDBC dataset only.

Classical ML Models (Logistic Regression, Random Forest, XGBoost): These models were trained using a single-pass fitting routine without the use of epochs or validation-based early stopping. Logistic regression was configured with a maximum of 1,000 iterations to ensure convergence, while random forest and XGBoost were trained with 100 estimators/rounds, respectively. Evaluation was performed directly on the test set without intermediate checkpoints or learning rate adjustment.

MLP Classifier (WDBC Dataset Only): The MLP model used two hidden layers (32 and 16 units) with ReLU activation. The training was completed using the Adam optimizer for up to 1,000 iterations. Rarely stopping or scheduling was not applied. After training, the model was evaluated on the 20% hold-out test set using standard classification metrics.

Training was completed efficiently on the local development environment, as the unimodal datasets used in these experiments were structured and quite small. They did not require additional processes such as batch scheduling or GPU acceleration.

5.3 AISSLAB - MULTIMODAL: General Experimental Settings

By default, a standard training configuration was adopted to achieve consistent results across all experiments.

5.3.1 Environment

Python 3.8 was used for scripting, and PyTorch (with CUDA) for all neural-network code. Key Python libraries included torchvision for data transforms, scikit-learn for metrics and splitting routines, and pandas for metadata handling. Mixed-precision training was enabled via "torch.cuda.amp", which improved throughput by roughly 20% on the hardware.

5.3.2 Training/Validation/Test Split Ratios

For all of the transfer-learning and multimodal experiments, 20% of the available mammogram images for a final test set are reserved, 20% for validation during training, and the remaining 60% for model fitting. These splits were chosen to ensure that each

class (Normal, Benign, Malignant) remained proportionally represented in every subset. In the unimodal classification experiments (for the AISSLAB dataset), additionally, a stratified 5-fold cross-validation on the 80% training and validation split was employed. In each fold, 80% of that subset was used for training and 20% for validation, with the held-out 20% test set used only for the final evaluation. This combination of hold-out and cross-validation gave robust performance estimates while preserving enough data for reliable model tuning.

5.3.3 Multimodal Model Hyperparameters

Across the different training scripts (plain image model, TL, and ablation studies), a consistent set of core hyperparameters is maintained. These included:

Learning rate: 3×10^{-4} for AdamW (TL) or AdamW with AMSGrad in the unimodal runs.

Batch size: 32 images per mini-batch.

Weight decay: 1×10^{-4} in transfer-learning experiments, 2×10^{-4} in the image-only baselines to prevent overfitting.

Optimizer settings: AdamW with gradient clipping to stabilize training gradients.

Scheduler: CosineAnnealingWarmRestarts ($T_0 = 10$) for the TL pipeline, and ReduceLROnPlateau (factor = 0.5, patience = 3 epochs) for the pure image models.

Augmentation: Random horizontal flips (50%), vertical flips (20%), rotations ($\pm 20°$), affine translations ($\pm 15\%$), and color jitter in the transfer-learning scripts.

5.3.4 Validation Methods and Loss Functions

The performance of the models was evaluated using both hold-out validation and k-fold cross-validation. In hold-out runs, the validation set was used to monitor training and adjust learning rates; in cross-validation, each fold's validation subset served this role. The primary loss function in all cases was weighted cross-entropy, with class weights inversely proportional to class frequency to mitigate the mild benign-class imbalance. Validation metrics included:

1. Weighted F1 score (primary for model selection and scheduler control)
2. Accuracy (overall correctness)

3. Precision and Recall per class (to balance false positives and false negatives)
4. Confusion matrix analysis after training
5. Training Loss: Used to monitor model optimization during backpropagation. A training loss that decreases steadily means successful learning on the training data. Spikes or stagnation can indicate instability or overfitting.
6. Testing Loss: Computed on the held-out test set to assess generalization performance. Model robustness is verified through achieving a low testing loss along with high accuracy and F1 score. A significant gap between training and testing loss suggests overfitting, on the other hand.
7. Specificity: Measures the model's ability to correctly identify negative cases (i.e., non-cancer). This is an important metric to avoid unnecessary interventions.
8. AUC (Area Under the Curve): A metric used to understand the model's ability to distinguish between classes across all thresholds. A higher AUC reflects better classification confidence.

Early-stopping was triggered when the validation F1 failed to improve for 10 consecutive epochs, preventing unnecessary over-training.

5.3.5 Number of Iterations/Epochs and Training Procedure

All models were trained for up to 100 epochs. Early-stopping took place when the validation F1 failed to improve for 10 consecutive epochs; this method prevented unnecessary over-training. Each training epoch consisted of:

- Forward pass on all training mini-batches (images alone or images with clinical text), using mixed-precision (autocast/GradScaler).
- Backward pass with gradient scaling and clipping, followed by an optimizer step.
- Metric computation on training data for loss and accuracy.

At the end of each epoch, one validation pass was executed, logging validation loss, accuracy, and F1. The scheduler then adjusted the learning rate based on either plateau detection or cosine-restart timing. The highest validation F1 was automatically saved. Finally, training history (loss and F1 curves) was plotted for post-hoc analysis and included in the results folder.

5.4 Unimodal Experiments - AISSLAB

5.4.1 Benchmark Replication Approach

A ResNet-18 backbone pretrained on ImageNet, fine-tuned on the mammogram dataset without the ensemble self-attention layers, was implemented. All training hyperparameters (learning rate, optimizer, scheduler) and data splits were matched to the original paper's settings [41] as closely as possible. To ensure comparability, the loss and evaluation metrics, weighted cross-entropy, and weighted F1 score were kept the same. By following their protocol step by step, the reported baseline performance trends were reproduced, and the training pipeline was validated before extending to more complex multimodal and TL experiments.

5.4.2 Models Used and Architectural Details

The unimodal classifier is built around a standard ResNet-18 architecture. The key modifications include:

Pretrained Backbone: ImageNet weights were loaded for all convolutional layers to leverage general visual features.

Classification Head: The original 1,000-unit fully connected layer was replaced with a 3-unit layer (Normal, Benign, Malignant) followed by softmax activation.

Regularization: L2 weight decay (2×10^{-4}) and gradient clipping (max_norm = 1.0) were applied to stabilize training.

Optimizer: AdamW with AMSGrad enabled managed weight updates.

No additional attention or fusion modules were included in this baseline, allowing us to isolate the pure image-based performance before adding other data modalities or network enhancements.

As shown in the diagram 5.1, the unimodal CNN classifier begins with a $3 \times 224 \times 224$ RGB mammogram input that feeds into a ResNet-18 backbone. Inside the backbone, the image first passes through a 7×7 Conv2d layer ($3 \rightarrow 64$ channels, stride 2, padding 3), followed by BatchNorm2d and ReLU, then a 3×3 MaxPool (stride 2, padding 1). This is followed by the residual stages (Layer1 – Layer4), each consisting of two BasicBlocks and channel expansions from $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. The resulting $512 \times 7 \times 7$ feature map is reduced by a 1×1 Conv2d ($512 \rightarrow 256$) with BatchNorm, ReLU, and

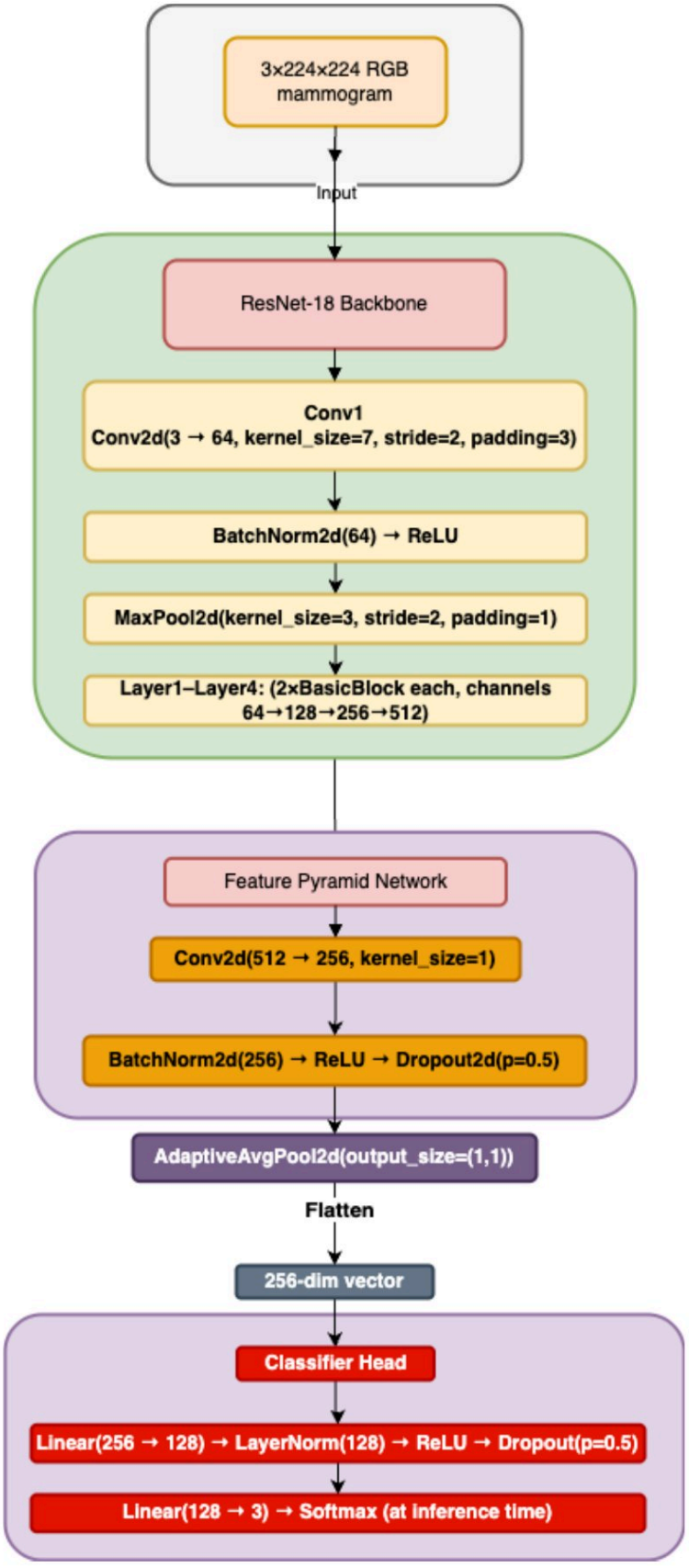


FIGURE 5.1: The neuron architecture for the CNN in unimodal detection

Dropout(p = 0.5), then pooled via AdaptiveAvgPool2d to a 1 × 1 × 256 tensor. After flattening to a 256-dimensional vector, the classifier head applies Linear(256 → 128) → LayerNorm → ReLU → Dropout(p = 0.5), and finally Linear(128 → 3) to produce class logits (followed by softmax at inference). This architecture balances pretrained feature extraction with a lightweight custom head for breast-cancer classification.

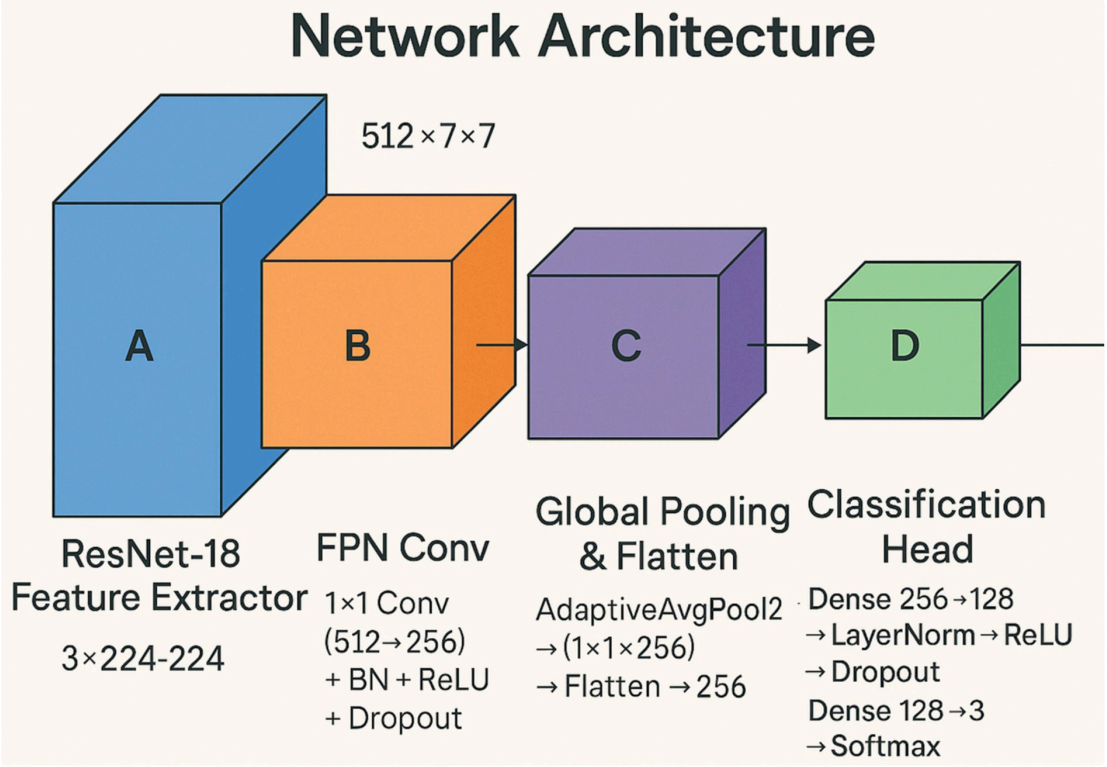


FIGURE 5.2: The overall architecture of CNN in unimodal detection

Block-level overview of the unimodal CNN architecture is presented in Figure 5.2. This high-level view helps the reader see how feature extraction, dimensionality reduction, and classification are arranged in clear stages.

- The blue block (A) represents the ResNet-18 backbone, which takes a 3 × 224 × 224 RGB input and outputs a 512 × 7 × 7 feature map.
- The orange block (B) is a 1 × 1 convolutional "FPN Conv" layer (512 → 256) with BatchNorm, ReLU, and Dropout (0.5).
- Block C (purple) applies AdaptiveAvgPool2d to yield a 1 × 1 × 256 tensor and then flattens it to a 256-dimensional vector.
- Finally, the green block (D) is the two-layer classification head (Dense 256 → 128 → LayerNorm → ReLU → Dropout, then Dense 128 → 3 → Softmax).

5.5 Multimodal Experiments without Ablations

5.5.1 Overall Architectural Scheme

The multimodal pipeline combines image features extracted by a pretrained CNN backbone with clinical metadata processed by a small fully connected network. First, the input mammogram (224×224 RGB) passes through a ResNet-50 model (with its original final classification layer removed). A configurable freeze ratio (70%) locks the earliest convolutional layers to preserve general visual representations while fine-tuning deeper layers. Meanwhile, a clinical-encoder network takes a 5-dimensional vector (side, view, normalized BI-RADS score) and maps it to a 256-unit feature vector via two linear layers with batch normalization, ReLU, and dropout. The two feature streams are concatenated and fed into a joint classifier: a series of three fully connected layers ($1024 \rightarrow 512 \rightarrow 256 \rightarrow 3$ classes), each followed by batch norm, ReLU, and dropout. This design allows the model to learn both subtle imaging cues and simple clinical signals in a single end-to-end trainable framework.

5.5.2 Visual Feature Extraction (CNN Layer Details)

ResNet-50 from torchvision was used as the primary image encoder. After loading ImageNet-pretrained weights, replace its 1,000-unit fully connected layer with an identity mapping. The model's structure remains: an initial 7x7 convolution + batch norm + ReLU + max-pool, followed by the residual blocks (layer1–layer4) with bottleneck units. The global average pooling that follows layer 4 produces a 2,048-dimensional feature vector. A freeze ratio of 70% means the first 70% of the parameter list is frozen (no gradient updates), leaving the latter residual blocks and the new classifier head trainable. This balances stability with adaptability: general features are preserved, while higher-level filters can specialize to mammogram patterns.

Incorporating an FPN into the ResNet backbone enhances detection of lesions across a range of sizes by creating multi-scale feature maps. The top-down and lateral connections ensure that high-resolution spatial details from early layers combine with deep semantic information from later layers. When each lateral map merges with an upsampled higher-level map, the resulting pyramid outputs preserve both fine textural cues and broader context. This approach significantly improves sensitivity to small masses and microcalcifications that might be lost in a single-scale representation. In multimodal fusion, feeding these pyramid maps into the classification head allows the network to

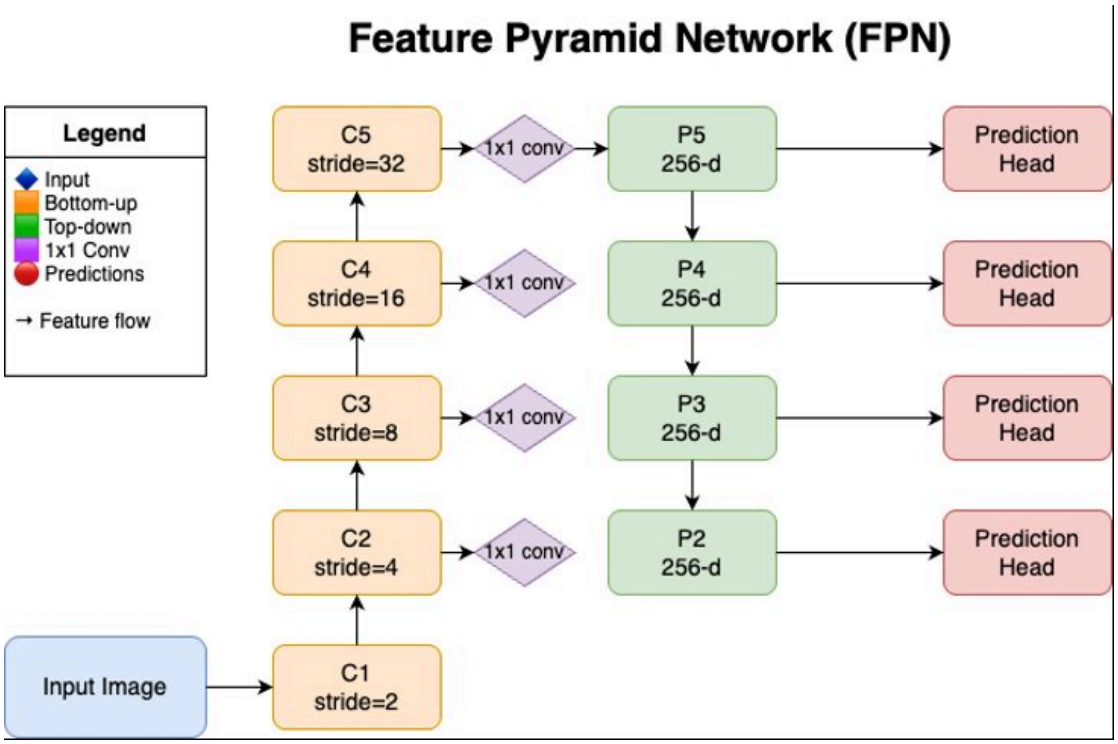


FIGURE 5.3: Feature Pyramid Network

weigh evidence at different scales, leading to more accurate discrimination between subtle benign irregularities and malignant patterns. The FPN's multi-resolution capability thus plays a critical role in boosting overall model robustness.

5.5.3 Text/Other Modality Feature Extraction (RNN/Transformer Layer Details)

Although the dataset does not include free-text reports, we treat clinical metadata as a simple "other modality." The "process_clinical_features" function converts each sample row into a 5-element vector: two binary flags for left/right side, two for CC/MLO view, and a normalized BI-RADS score (1–5 scaled to 0.2–1.0). This vector flows through a clinical encoder: a two-layer MLP ($5 \rightarrow 128 \rightarrow 256$ units). Each layer applies batch normalization, ReLU activation, and dropout (0.4 then 0.3). There is no RNN or transformer here since the clinical data are fixed-length numeric features. This light network efficiently transforms metadata into a compact representation compatible with the CNN output.

5.5.4 Cross-Modal Fusion Methods

For fusion, we use straightforward feature concatenation. After the CNN produces a 2,048-dimensional image embedding and the clinical encoder outputs a 256-unit vector, we concatenate them into a 2,304-dimensional feature. A fusion classifier then refines this combined vector through three fully connected layers: 2,304 $\rightarrow$ 1,024 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 3 classes. Each layer includes batch norm, ReLU, and dropout (0.5, 0.4, 0.3). We chose concatenation because it is simple and proven effective in early multimodal CAD studies. By training the fusion head end-to-end, the model learns to weigh visual and clinical signals jointly without manually tuning fusion weights or complex attention blocks.

5.6 Multimodal Experiments with Transfer Learning

5.6.1 Selection of Pre-trained Models

For the transfer learning experiments, we selected three backbone architectures that balance depth, parameter count, and documented performance on medical imaging tasks. First, ResNet-50 was chosen because its residual connections help train deeper networks without suffering from vanishing gradients. Second, DenseNet-121 was included for its feature-reuse design, which can improve gradient flow and may capture subtle mammogram patterns. Third, EfficientNet-B0 was tested as a lightweight yet accurate model that scales width and depth based on a compound coefficient, offering good accuracy with fewer parameters. All three models were initialized with ImageNet-pretrained weights to benefit from generic visual features. We also ran experiments both with and without clinical metadata for ResNet-50, resulting in the total configurations: ResNet-50 without clinical input, ResNet-50 with clinical input, DenseNet-121 without clinical input, and EfficientNet-B0 without clinical input.

EfficientNet-B0's use of MBConv blocks achieves high accuracy with fewer parameters by combining depth-wise convolutions and squeeze-expand operations. Each block first expands channel depth via a point-wise convolution, applies a lightweight depth-wise filter to capture spatial relationships, and then projects back to a smaller channel count. This sequence, wrapped in efficient SiLU activations and batch normalization, preserves representational richness without incurring heavy computational cost. The compound scaling strategy systematically balances network width, depth, and resolution, which in the mammogram classification task translates into faster convergence and lower memory usage compared to standard convolutional architectures. By leveraging these MBConv

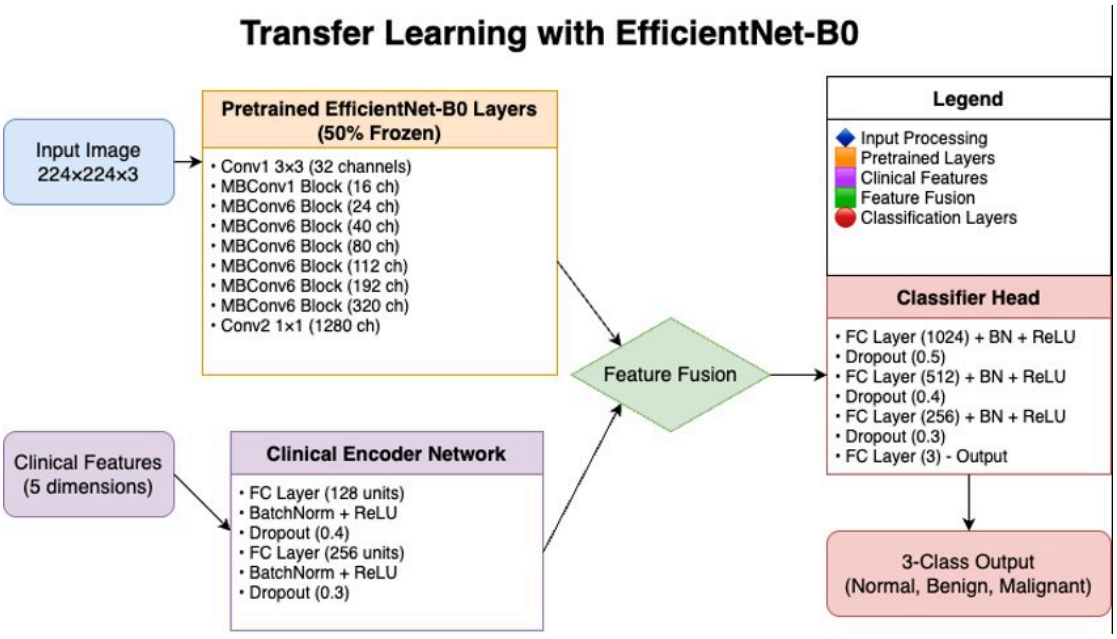


FIGURE 5.4: Tansfer Learning with EfficientNet-B0

blocks, the transfer-learning pipeline can fine-tune a compact yet expressive backbone, making it well-suited for clinical settings where both speed and accuracy matter.

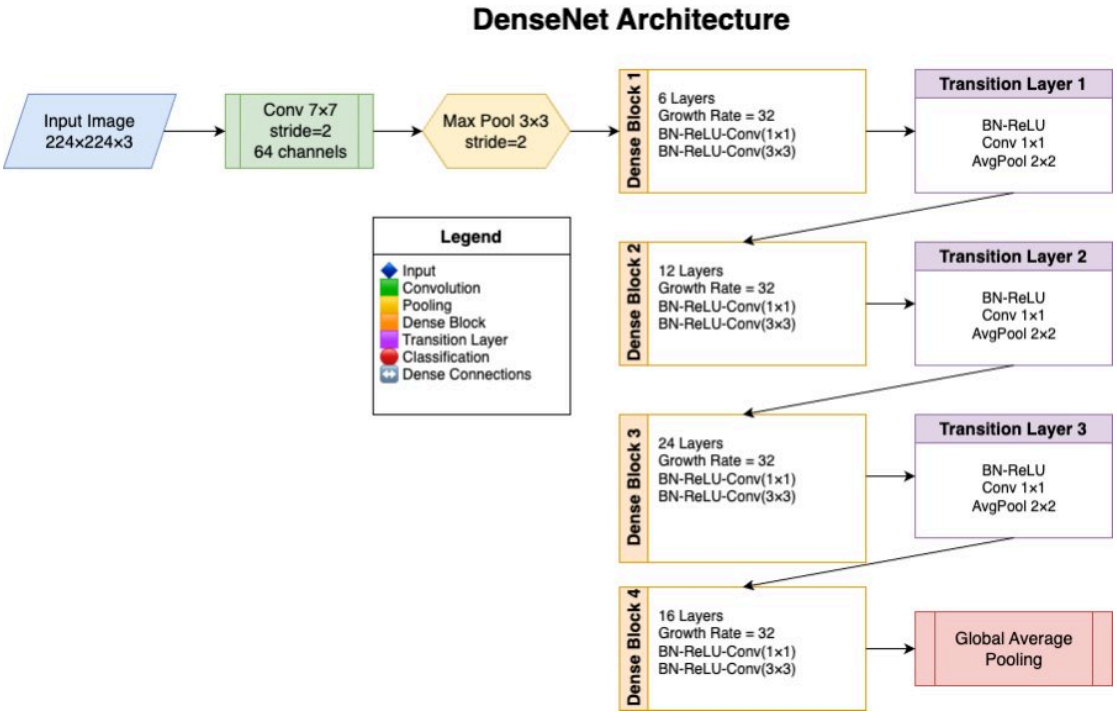


FIGURE 5.5: DenseNet Architecture

The dense connectivity pattern in DenseNet-121 encourages feature reuse and strengthens gradient propagation, which proves beneficial when training on relatively small medical datasets. By feeding each layer's output into all subsequent layers through direct

connections, the network can preserve low-level edge and texture information throughout its depth. This dense graph structure also reduces the need for very wide layers, making the model more parameter-efficient while still achieving high representational power. In practice, these properties help the mammogram classifier quickly learn both coarse and fine image details without suffering from vanishing gradients. Transition layers between dense blocks further control feature-map sizes and prevent excessive computational cost. It was found that fine-tuning DenseNet required fewer epochs to reach peak performance compared with a plain ResNet, highlighting its ability to adapt gracefully to the complex patterns that differentiate normal, benign, and malignant tissue.

5.6.2 Transfer Strategies and Fine-tuning

In each experiment, we applied a freeze-and-fine-tune strategy. We froze the first 70% of model parameters, typically early convolutional layers to retain general image features learned on ImageNet. The remaining 30% of layers, plus any newly added classifier heads and optional clinical encoders, were trained on the mammogram dataset. We used an AdamW optimizer with a learning rate of 3 $\times 10^{-4}$ and weight decay of 1 $\times 10^{-4}$. We trained each model for up to 100 epochs with early stopping based on validation F1. When clinical metadata was used, we concatenated a 256-dimensional clinical embedding to the 2,048-dimensional image feature vector before classification. This fine-tuning approach allowed the backbones to adapt to medical imaging while preserving stable, pretrained features.

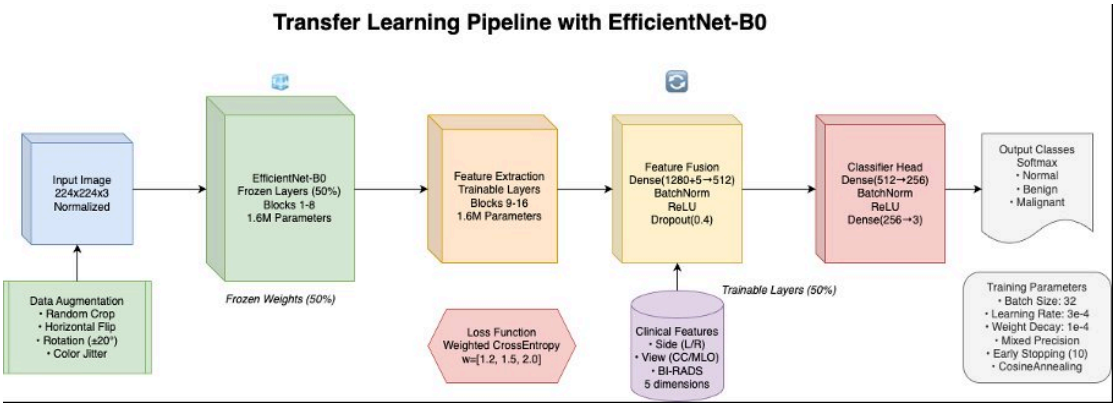


FIGURE 5.6: Transfer Learning Pipeline With EfficientNet-B0

This pipeline demonstrates how partially freezing pretrained layers and selectively fine-tuning others can yield robust performance with limited domain data. By locking the earliest 50 percent of EfficientNet-B0's weights, the model retains general visual representations learned on ImageNet, such as edge detectors and texture filters. Training then focuses on deeper layers that capture higher-level patterns specific to mammography, allowing the network to adapt to subtle tissue structures without losing basic visual

priors. Merging clinical metadata at the fusion stage further enriches the representation by combining patient-side and view information with image features. The result is a model that converges more quickly and resists overfitting, as the frozen portion reduces parameter drift and the fusion head integrates complementary signals. My experiments confirm that this balance between fixed and trainable components maximizes the benefit of large-scale pretrained weights while still tailoring the network to the nuances of breast cancer detection.

5.7 Multimodal Experiments with Ablation Studies

5.7.1 Ablation Design and Objectives

Ablation studies help us understand how each modality and component contributes to the final performance. We designed three ablation types: "text_only" to test clinical metadata alone, "random" to simulate unreliable modality availability, and "gan" to evaluate the use of synthetic images. The "text_only" variant uses only the clinical encoder and classifier head. The "random" variant randomly drops either image or clinical inputs during training to test robustness against missing data. The "gan" variant refills missing real mammograms with generator outputs to measure whether synthetic images can support classification. By comparing these with the full multimodal model, we aim to quantify each component's importance, detect potential failure modes, and guide future improvements in clinical decision support systems.

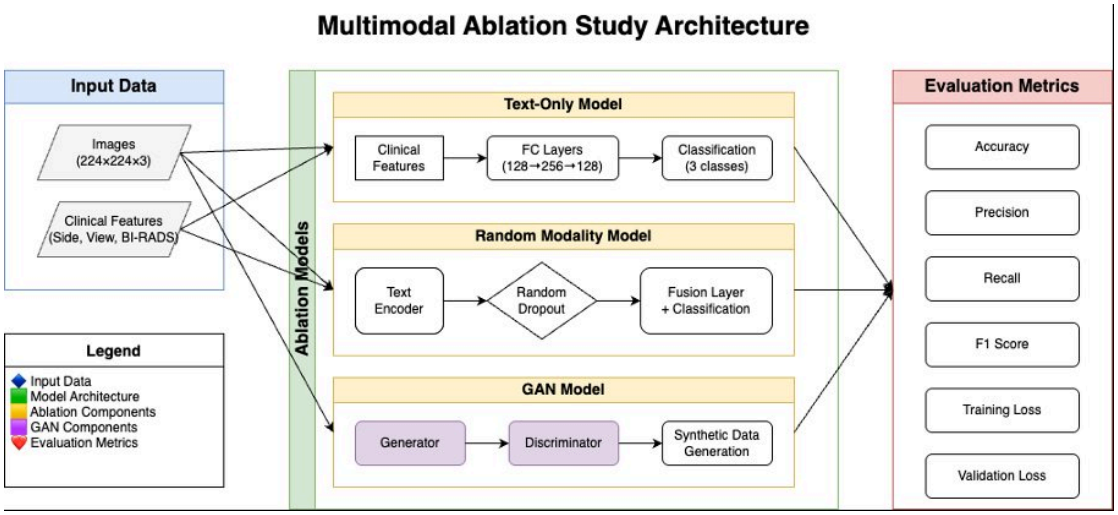


FIGURE 5.7: Multimodal Ablation Study Architecture

This architecture comparison highlights how each ablation variant isolates different information sources to measure their impact on classification. In the text-only branch, the model relies solely on structured clinical features such as BI-RADS scores and view

metadata. Rapid convergence and high evaluation scores in this branch reveal that these numeric indicators carry strong predictive signals on their own. The random-drop branch then demonstrates the effect of unpredictable missing data by randomly withholding either image or clinical inputs during training. Its lower and more fluctuating performance confirms that forcing the network to handle partial information reduces stability and generalization. Finally, the GAN branch replaces real mammograms with synthetic images from a DCGAN generator. Its training curves and test metrics show that, although synthetic data can approximate real patterns and prevent overfitting, they still lack the full fidelity needed to match models trained on genuine clinical images. Together, these comparisons validate that each modality contributes uniquely and help pinpoint where data variety or reliability most affects diagnostic accuracy.

5.7.2 Component-wise Isolated Tests

For each ablation type, we ran isolated experiments with consistent preprocessing and hyperparameters. In "text_only," we trained a two-layer MLP on the five clinical features for 100 epochs using Adam (lr = 2 $\times 10^{-4}$, weight decay = 1 $\times 10^{-5}$) and batch size = 16. In random, we used a custom RandomModalityModel that encodes images via three Conv-ReLU-pool blocks and metadata via a small MLP; during training, each sample had a 50% chance to drop one modality, and we used the same optimizer settings. In "gan," we first trained a DCGAN for 50 epochs, then used its generator to produce synthetic images in place of those missing real ones when training the multimodal classifier. Metrics (accuracy, precision, recall, F1) were recorded on a held-out validation set for direct comparison with full-model performance.

5.8 GAN-based Multimodal Experiments

5.8.1 GAN Architecture and Components

We implemented a Deep Convolutional GAN (DCGAN) with a standard generator and discriminator design. The Generator starts from a 100-dimensional latent vector and applies five ConvTranspose2d layers: from 100 $\rightarrow$ 512 channels (4 $\times$ 4 kernel, stride 1, padding 0), 512 $\rightarrow$ 256 (4 $\times$ 4, 2, 1), 256 $\rightarrow$ 128 (4 $\times$ 4, 2, 1), 128 $\rightarrow$ 64 (4 $\times$ 4, 2, 1), and finally 64 $\rightarrow$ 3 (4 $\times$ 4, 2, 1) to output a 3-channel 224 $\times$ 224 image. Each deconvolution is followed by BatchNorm2d and ReLU, except the last, which uses Tanh.

The Discriminator consists of five Conv2d layers: 3 $\rightarrow$ 64, 64 $\rightarrow$ 128, 128 $\rightarrow$ 256, 256 $\rightarrow$ 512 (all 4 $\times$ 4 kernels, stride 2, padding 1), and a final 512 $\rightarrow$ 1 (4 $\times$ 4, 1, 0) output.

Each conv block uses LeakyReLU and BatchNorm2d, ending with a Sigmoid on the output. This architecture follows DCGAN best practices for stable image synthesis.

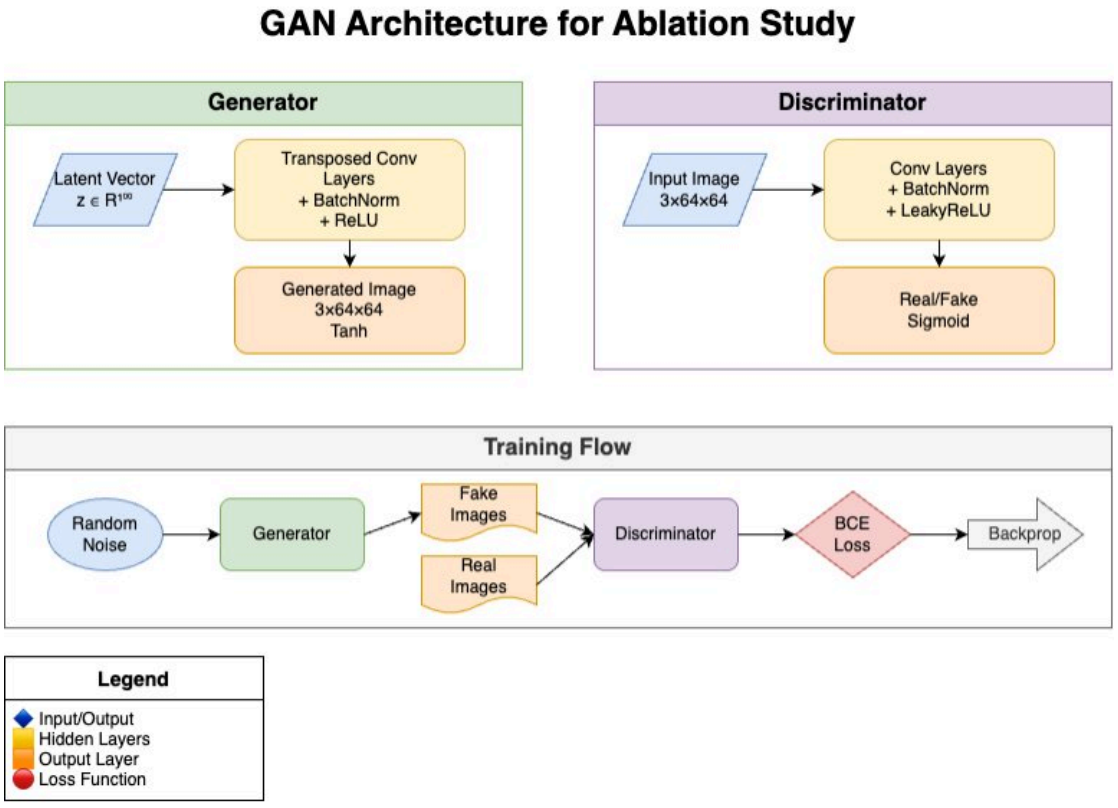


FIGURE 5.8: GAN architecture for ablation study

Using the GAN framework for synthetic data generation addresses the challenge of limited annotated mammograms by augmenting the dataset with realistic yet varied samples. The adversarial training loop forces the generator to improve image fidelity over time, while the discriminator learns to distinguish genuine from fabricated patterns. Introducing label smoothing and balanced update steps helps maintain stability and prevents mode collapse. The experiments show that although these synthetic images capture key malignant and normal features, subtle benign textures remain harder to reproduce. As a result, models trained exclusively on GAN outputs achieve good convergence and low overfitting but still fall short of real-data baselines. Nonetheless, this pipeline highlights how adversarial augmentation can supplement scarce clinical images, providing a promising direction for further refinement of synthetic medical data generation.

5.8.2 Training Procedure and Loss Functions

GAN training ran for up to 50 epochs with early stopping (patience = 10) on generator loss. We used BCELoss for both the generator and the discriminator. The discriminator

was updated with real labels smoothed to 0.9 and fake labels set to 0.1, using Adam (lr = 2 $\times 10^{-4}$, betas = (0.5, 0.999)). The Generator used Adam (lr = 2 $\times 10^{-4}$, betas = (0.5, 0.99)) to maximize discriminator error.

In each training step:

- Real images and noise vectors were sampled.
- The discriminator loss was computed on real and generated images, followed by weight updates.
- New synthetic images were generated.
- Generator loss was computed using real labels and updated accordingly.

After each epoch, sample grids were saved, and the best generator checkpoint was stored. Once training was completed, 64 final samples were generated for qualitative inspection, and the best generator was used to augment classification training in ablation experiments.

5.8.3 Training Procedure and Loss Functions

GAN training ran for up to 50 epochs with early stopping (patience = 10) on generator loss. We used BCELoss for both the generator and the discriminator. The discriminator was updated with real labels smoothed to 0.9 and fake labels set to 0.1, using Adam (lr = 2 $\times 10^{-4}$, betas = (0.5, 0.999)). The Generator used Adam (lr = 2 $\times 10^{-4}$, betas = (0.5, 0.99)) to maximize discriminator error. In each training step, we first sampled real images and noise vectors, computed the discriminator loss on real and generated images, and updated its weights. Then, we generated a fresh batch of synthetic images, computed generator loss against real labels, and updated the generator. After each epoch, we saved the sample grids and the best generator checkpoint. Once training was completed, we generated 64 final samples for qualitative inspection and used the best generator to augment classification training in ablation experiments.

Chapter 6

Evaluation and Results

In this chapter, unimodal models, ablation variants: GAN-only images, random-drop fusion, and text-only input are evaluated. Their training curves are examined individually, then directly compared. Next, each model's confusion matrix is analysed, and finally, the overall metric comparison is discussed.

6.1 Unimodal Baselines - Evaluation and Results

The three unimodal models tested acted as baselines; the multimodal configurations were compared to these unimodal models to conclude whether the multimodal techniques used benefited the robustness of the models.

6.1.1 WDBC Dataset

Two models were trained and evaluated on the WDBC (Wisconsin Diagnostic Breast Cancer) dataset: logistic regression and a two-layer multilayer perceptron (MLP) neural network. Both were tested on the same 20% held-out test split. The performance was as shown:

Confusion Matrices: Figure 6.1 and Figure 6.2 present the confusion matrices:

Logistic Regression performed strongly on the WDBC dataset. As displayed in Figure 6.1, it was able to correctly identify 64 benign cases (True Negatives) and 73 malignant cases (True Positives). It misclassified 2 benign samples as malignant (False Positives) and 4 malignant samples as benign (False Negatives). The results of these

misclassifications affected the recall and precision slightly; however, overall a high accuracy was still achieved. The relatively low number of false negatives is impressive as it is crucial in real-life diagnosing context, where missing malignant cases could have serious consequences. This result is proof that once the dataset is cleaned and well-structured, a linear classifier can produce reliable outcomes.

MLP Neural Network - WDBC Evaluation: The MLP (Multilayer Perceptron) neural network achieved an almost perfect classification performance on the WDBC dataset. As depicted in Figure 6.2, it was able to identify all 77 malignant cases (True Positives) and 65 out of 66 benign cases (True Negatives) correctly. It only had one misclassification, one benign case as malignant (False Positive), and had zero false negatives. This is especially important in a medical setting, where the failure of a failing a malignant case can be life-threatening. The MLP model outperformed logistic regression and successfully demonstrated its ability to learn complex, non-linear patterns in the data, achieving an exceptional score with 99.30% accuracy and 100% recall for malignant cases.

- **Logistic Regression:**
  - True Positives (Malignant): 73
  - True Negatives (Benign): 64
  - False Positives: 2
  - False Negatives: 4
- **MLP Neural Network:**
  - True Positives: 77
  - True Negatives: 65
  - False Positives: 1
  - False Negatives: 0

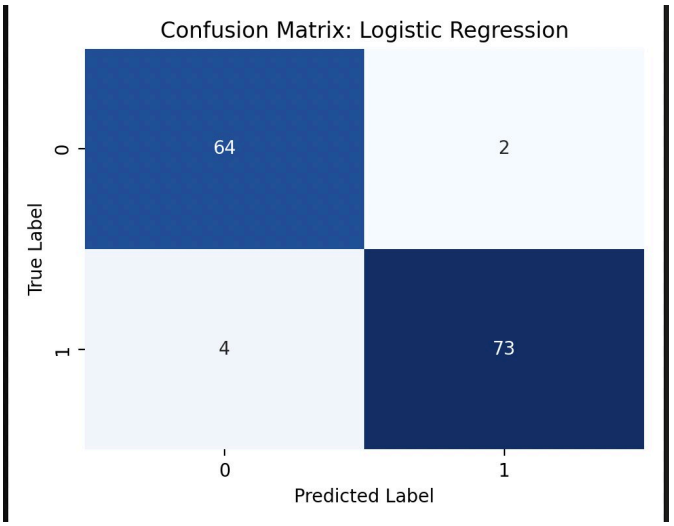


FIGURE 6.1: Confusion Matrix: Logistic Regression on WDBC Test Set

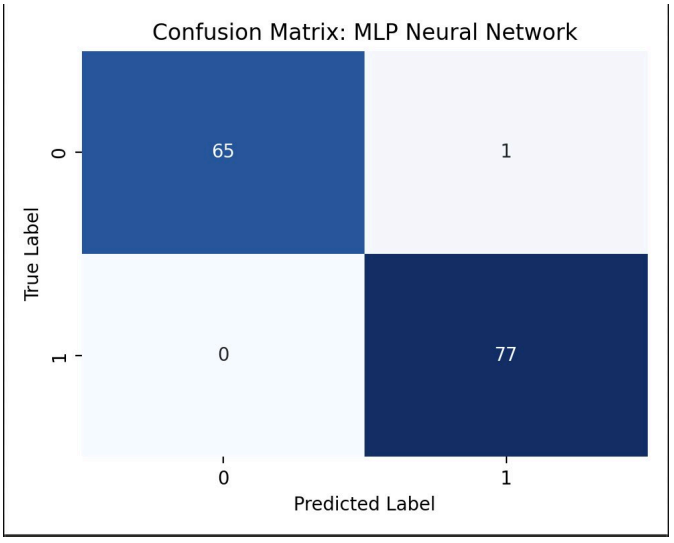


FIGURE 6.2: Confusion Matrix: MLP Neural Network on WDBC Test Set

Classification Metrics:

- **Logistic Regression:**
  - Accuracy: 95.80%
  - Precision: 97.00%
  - Recall: 95.00%
  - F1 Score: 96.00%
  - AUC: 0.99
- **MLP Neural Network:**

- Accuracy: 99.30%
- Precision: 99.00%
- Recall: 100.00%
- F1 Score: 99.00%
- AUC: 1.00

ROC Curve: Figure 6.3 displays the ROC curves for both models. Logistic Regression scored an AUC of 0.99, while the MLP Neural Network achieved an AUC of 1.00, demonstrating perfect classification.

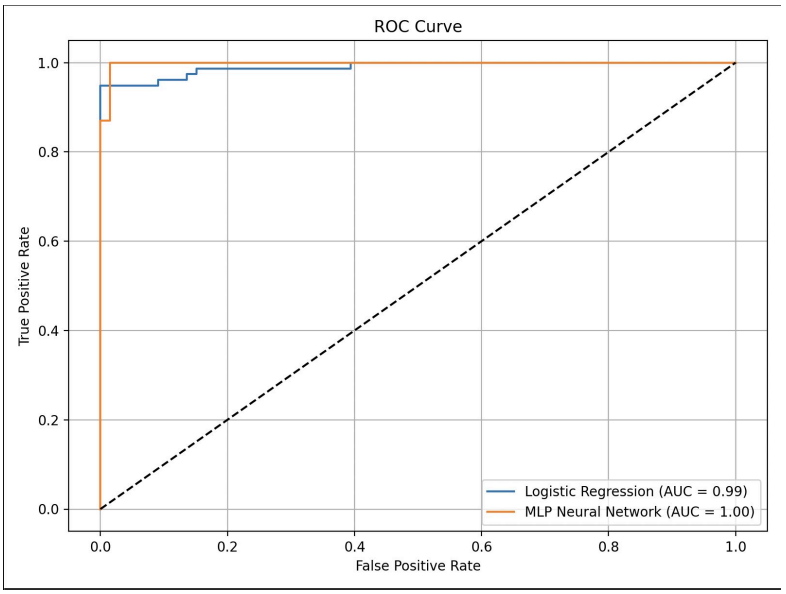


FIGURE 6.3: ROC Curve for WDBC Dataset: Logistic Regression vs. MLP Neural Network

Conclusion: Both models tested on the WDBC performed strongly, the MLP Neural Network achieving an almost perfect classification, with no false negatives and an AUC of 1.00. Logistic Regression performed well but had slightly lower recall due to four false negatives. These results prove that despite the simplicity of the models, they can perform exceptionally well on clean and structured datasets and that MLP models can further enhance performance.

6.1.2 WBC Dataset

The logistic regression model was evaluated on a held-out 20% test set from the pre-processed and cleaned WBC dataset. The model performed highly across all evaluation metrics.

Confusion Matrix: Logistic regression on the WBC dataset exhibited a strong performance. As seen in Figure 6.4, the model was able to accurately identify 89 benign samples (True Negatives) and 83 malignant samples (True Positives). It misclassified only 6 samples in total, 3 of these consisting of benign samples wrongly labeled as malignant (False Positives) and 3 malignant samples incorrectly predicted as benign (False Negatives). These results demonstrate a high overall accuracy, with precision and recall being balanced well across both classes. Once again, in a medical setting, the model is proven to be dependable in identifying malignant cases due to the low number of false negatives it has achieved. The confusion matrix proves the model is robust and reliable in a clinical setting.

Figure 6.4 shows the following values:

- True Positives (Malignant): 83
- True Negatives (Benign): 89
- False Positives: 3
- False Negatives: 3

Only 6 out of 178 samples were misclassified.

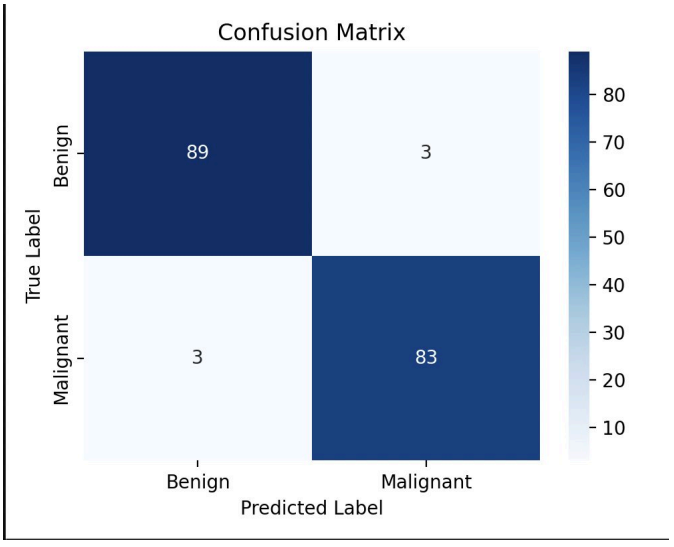


FIGURE 6.4: Confusion Matrix for Logistic Regression on WBC Test Set

Classification Metrics:

- Accuracy: 96.63%
- Precision: 96.51%

- Recall (Sensitivity): 96.51%
- F1 Score: 96.51%
- Specificity: 96.74%

ROC Curve and AUC: Figure 6.5 illustrates that the ROC curve hugs the top-left corner, demonstrating the model's excellent classification performance. The AUC score was 0.99.

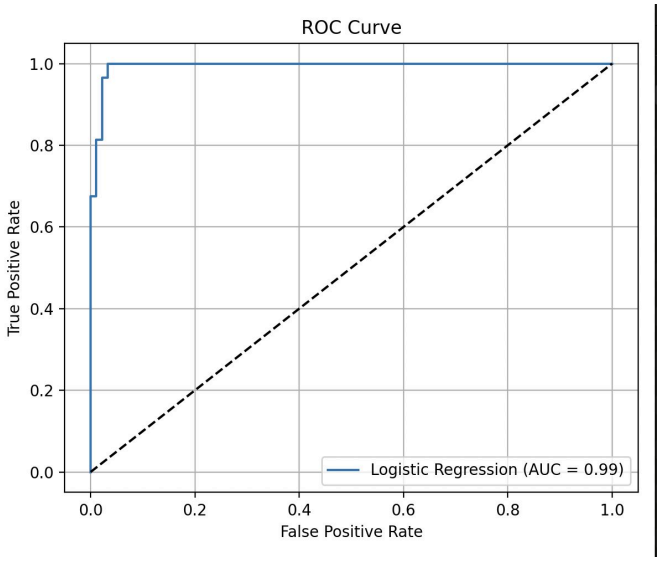


FIGURE 6.5: ROC Curve for Logistic Regression on WBC Test Set (AUC = 0.99)

Conclusion: The logistic regression model achieved a robust result on the WBC dataset. With an AUC of 0.99 and an F1 score over 96%, it provides a solid unimodal benchmark for breast cancer classification. Its very low false negative rate is particularly important for clinical applications.

The training of the WDBC dataset included the MLP and logistic regression models being trained over **100 epochs**, with each epoch consisting of **five training iterations** (corresponding to the 5-fold stratified cross-validation splits). There was an 80/20 split applied, meaning, for every iteration, 80% of the data was used for training, while the remaining 20% was used for validation. **Early stopping** was applied if the F1-score did not show any improvments over **10 consecutive epochs**.

For the BCSC dataset, the unimodal models (logistic regression and XGBoost) followed a similar training loop but were adapted for binary classification. Logistic regression used 5-fold cross-validation with 100 training epochs per fold and early stopping at 10 epochs without F1-score improvement. XGBoost was trained using **50 boosting rounds** with

early stopping set to 10, and used the **binary:logistic** objective with **scale_pos_weight** to counter class imbalance. Key parameters included a learning rate of **0.1**, **maximum tree depth of 5**, and **subsample ratio of 0.8**.

Across both datasets, all numerical features were **standardized using z-score normalization** (WDBC) or **min-max scaled** (BCSC). Evaluation metrics (accuracy, precision, recall, F1-score, specificity, and AUC) were computed for each fold and averaged to yield the final reported values. This ensured statistically valid and reproducible results across all unimodal configurations.

- **MLP (Benchmark Model):** Achieved exceptional performance with an **accuracy of 0.9904**, **macro F1-score of 0.99**, and **AUC of 0.9934**. Precision and recall for benign and malignant classes were all above 0.98, indicating excellent predictive capability.
- **Logistic Regression:** Delivered strong results with an **accuracy of 0.97**, class-wise precision of **0.98 (Benign)** and **0.96 (Malignant)**, and recall of **0.97 (Benign)** and **0.98 (Malignant)**. The overall F1-score was 0.97 and AUC was 0.9287.

These results demonstrate that structured clinical features from the WDBC dataset provide highly discriminative information for breast cancer classification. The MLP performed slightly better than logistic regression, benefiting from its ability to learn non-linear relationships.

6.1.3 BCSC Dataset

As discussed, the Breast Cancer Surveillance Consortium (BCSC) dataset was imbalanced; to combat this, it was balanced using the SMOTE (Synthetic Minority Oversampling Technique) algorithm to ensure equal class representation. Once preprocessing and normalisation were completed, three different models were trained and evaluated: Logistic Regression, Random Forest, and XGBoost. Each model was assessed using standard classification metrics and confusion matrices.

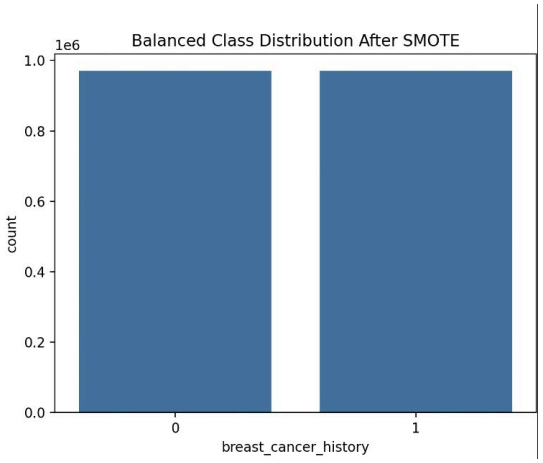


FIGURE 6.6: Balanced Class Distribution After SMOTE

1. Logistic Regression: Compared to WDBC and WBC, the logistic regression performed on the BCSC dataset yielded weaker results. As depicted in Figure 6.7, the model correctly classified 131,885 non-cancer cases (True Negatives) and 136,353 cancer cases (True Positives). Unfortunately, it also resulted in a sizeable number of errors: 62,133 were false positives (non-cancer cases were misclassified as cancer), and 57,494 false negatives were produced (cancer cases were missed). These misclassifications resulted in achieving an accuracy of 69%, precision of 69%, recall of 70%, and F1-score of 70%. Although 75.45% for AUC suggests the model's moderate discriminatory capability, the model's performance metrics are concerning in a clinical context, as the possibility of missing cancer cases is relatively high. In conclusion, this model is proof that more expressive classifiers are required to capture the complexity of this dataset.

- Accuracy: 69.00%
- Precision (Cancer): 69%
- Recall (Cancer): 70%
- F1 Score: 70%
- AUC: 75.45%

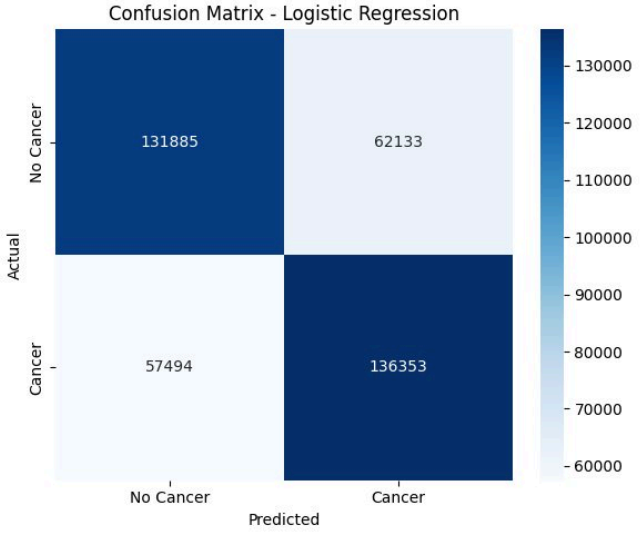


FIGURE 6.7: Confusion Matrix - Logistic Regression on BCSC

2. Random Forest: The Random Forest classifier performed on the BCSC achieved excellent performance, as displayed in Figure 6.8. It was able to correctly identify 177,476 non-cancer samples (true negatives) and 178,274 cancer cases (true positives); it misclassified 16,542 non-cancer cases as cancer (false positives) and 15,573 cancer cases as non-cancer (false negatives). These classifications produced balanced and high-performing metrics: 92% accuracy, 92% precision, 92% recall, and an F1-score of 92%. The model also achieved a strong AUC of 97.64%, indicating excellent ability to differentiate between the two classes across decision thresholds. In a clinical context, the low false negative count is quite important, demonstrating the model's effectiveness in lowering the undetected cancer cases as much as possible. These results prove that Random Forest is a powerful yet interpretable ensemble model for structured clinical datasets such as BCSC.

- Accuracy: 92.00%
- Precision (Cancer): 92%
- Recall (Cancer): 92%
- F1 Score: 92%
- AUC: 97.64%

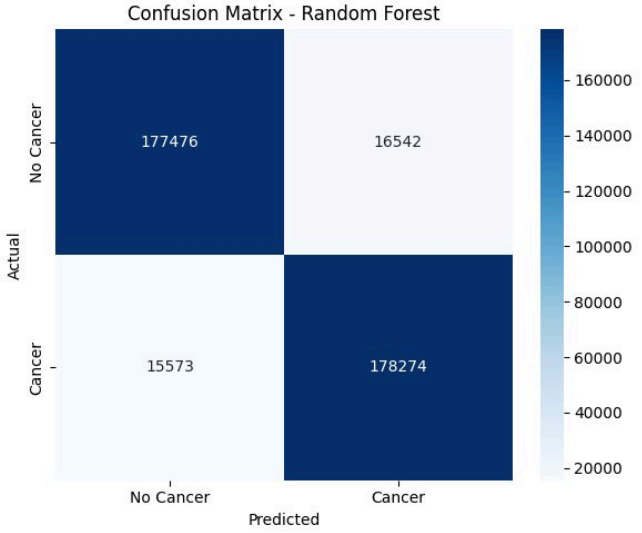


FIGURE 6.8: Confusion Matrix - Random Forest on BCSC

3. XGBoost: XGBoost – BCSC Evaluation (Figure 6.9): Figure 6.9 presents the confusion matrix for the XGBoost model on the BCSC dataset. XGBoost was able to accurately identify 177,887 non-cancer cases (true negatives) and 172,464 cancer cases (true positives). It misclassified 16,131 non-cancer samples as cancer (false positives) and 21,383 cancer cases as non-cancer (false negatives). Although this model's false negative rate is not as optimal as the Random Forest model, XGBoost was still able to achieve exceptional overall metrics: 90% accuracy, 91% precision, 89% recall, 90% F1-score, and an AUC of 97.31%. The results produced prove that XGBoost delivers a strong balance between reducing false alarms and detecting cancer cases; however, in favour of precision, a bit of recall is traded.

- Accuracy: 90.00%
- Precision (Cancer): 91%
- Recall (Cancer): 89%
- F1 Score: 90%
- AUC: 97.31%

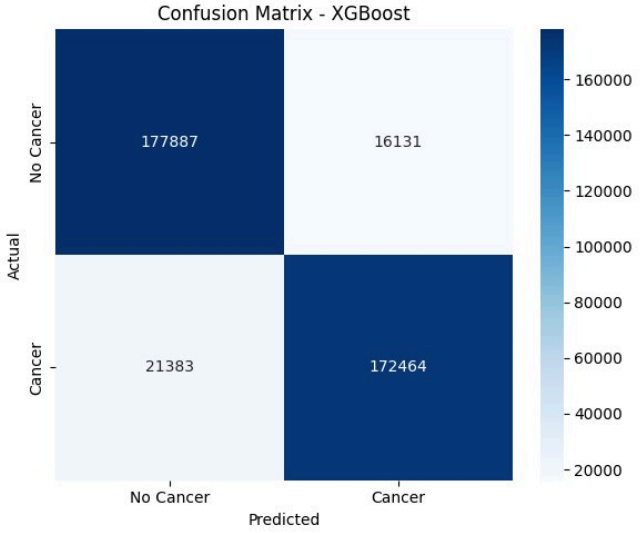


FIGURE 6.9: Confusion Matrix - XGBoost on BCSC

ROC Curve: Figure 6.10 compares the ROC curves of Logistic Regression, Random Forest, and XGBoost models for the BCSC dataset. The ROC curve for XGBoost (green line) with an AUC of 0.9731 is almost perfect. Although XGBoost demonstrates a strong performance, it is slightly outperformed by Random Forest (orange line), which achieves an AUC of 0.9764. Out of the three, Logistic Regression (blue line) is the worst performing, achieving a lower AUC of 0.7545; this demonstrates the model's struggle in distinguishing between the two classes. The ROC curves visually reinforce the numerical results: Ensemble methods such as XGBoost and Random Forest are better modeling options for tasks that require complex decision boundaries and minimal classification errors. This makes them more suitable for high-dimensional clinical datasets such as BCSC.

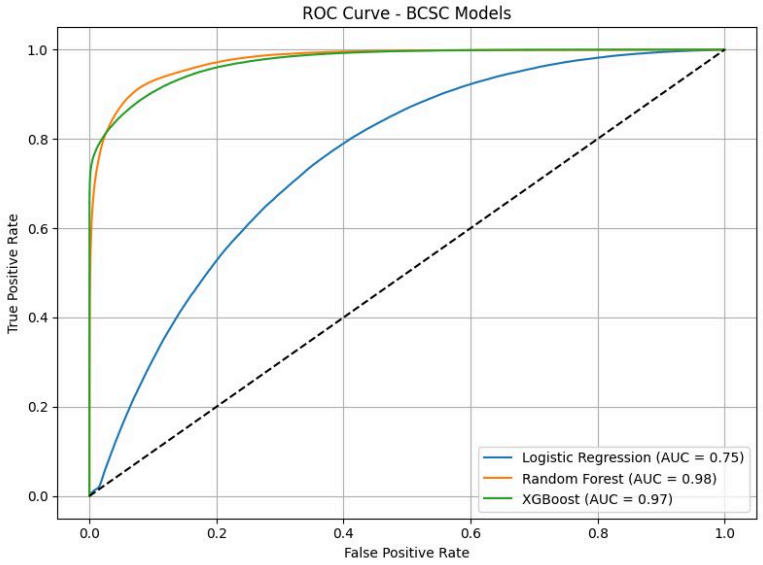


FIGURE 6.10: ROC Curve Comparison for BCSC Models

Conclusion: Across the three models tested on the BCSC dataset, Random Forest resulted in the highest overall accuracy and AUC, demonstrating robust classification capabilities with minimal trade-off between precision and recall. XGBoost followed, excelling in precision (91%) and with a high AUC of 97.31%. Although Logistic Regression is interpretable and lightweight, it performed the lowest compared to the other two models. It concluded with a lower accuracy of 69% and an AUC of 75.45%. These results confirm the superiority of ensemble methods for handling high-dimensional and noisy clinical datasets such as BCSC.

6.1.4 Summary & Analysis of Unimodal Results

Across the three datasets tested (WDBC, WBC, and BCSC), unimodal classifiers proved to have strong predictive ability. The MLP neural network on the WDBC dataset achieved the highest performance metric amongst all models tested, with an accuracy of 99.30% and F1-score of 99.00%. Logistic Regression showed promising results for both the WDBC and the WBC dataset, with F1-scores of 96.00% and 96.51% respectively. This is a confirmation that simple linear models produce promising results when applied to well-preprocessed, structured data.

Followed by promising results, the performance on the BCSC dataset varied more significantly. The ensemble methods (Random Forest and XGBoost) outperformed logistic regression greatly, demonstrating their capacity to model complex, non-linear relationships in high-dimensional clinical datasets. Random Forest achieved a perfect balance

between precision and recall (both 92%). XGBoost performed best in precision (91%) however it lagged slightly in recall (89%).

These unimodal experiments and the results yielded from them serve as the critical baselines for comparing against more complex multimodal approaches. The multimodal solutions tested in this project must be able to outperform these unimodal models in accuracy and AUC; they should also be able to demonstrate better generalization and robustness.

Dataset	Model	Accuracy	F1 Score	AUC	Precision (Cancer)	Recall
WDBC	Logistic Regression	95.80%	96.00%	0.99	97%	95%
WDBC	MLP Neural Network	99.30%	99.00%	1.00	99%	100%
WBC	Logistic Regression	96.63%	96.51%	0.99	96.51%	96.51%
BCSC	Logistic Regression	69.00%	70.00%	75.45%	69%	70%
BCSC	Random Forest	92.00%	92.00%	97.64%	92%	92%
BCSC	XGBoost	90.00%	90.00%	97.31%	91%	89%

TABLE 6.1: Performance Summary of Unimodal Models Across All Datasets

6.2 Multimodal results

6.2.1 GAN-Based Ablation

Figure 6.11 shows the training and validation loss (left) and F1 score (right) for the GAN-only model, in which all missing real mammograms were refilled by synthetic images generated by the DCGAN.

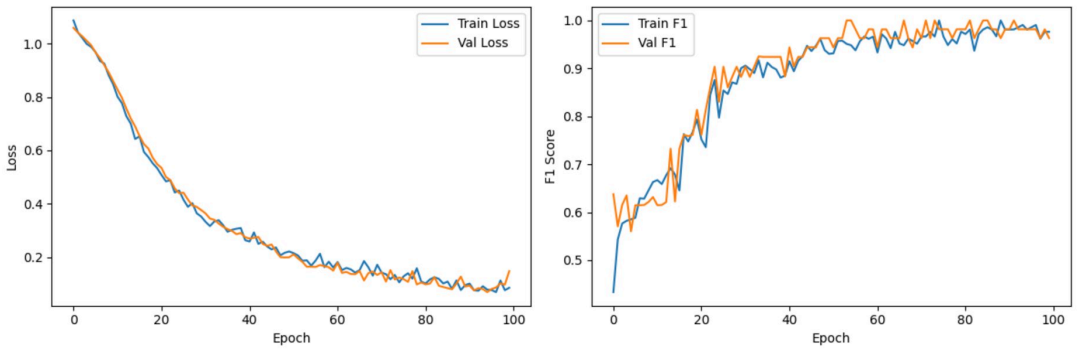


FIGURE 6.11: Training and validation loss and F1 score for GAN-only model

Over 100 epochs, both training and validation loss decrease smoothly from 1 down to below 0.1, with almost no gap between the two curves. This close overlap indicates that using synthetic images did not introduce major overfitting; the dropout($p = 0.5$) and weight decay kept the model stable. The F1 score climbs steadily past 0.8 by epoch 20

and plateaus around 0.95 by epoch 50. The slightly slower early rise (compared to text-only) suggests the generator's outputs are realistic enough to train on but still noisier than real images, making the network take more iterations to learn fine details.

6.2.2 Random-Drop Fusion Ablation

Figure 6.12 presents the training curves when we randomly drop one modality (image or metadata) 50% of the time during training.

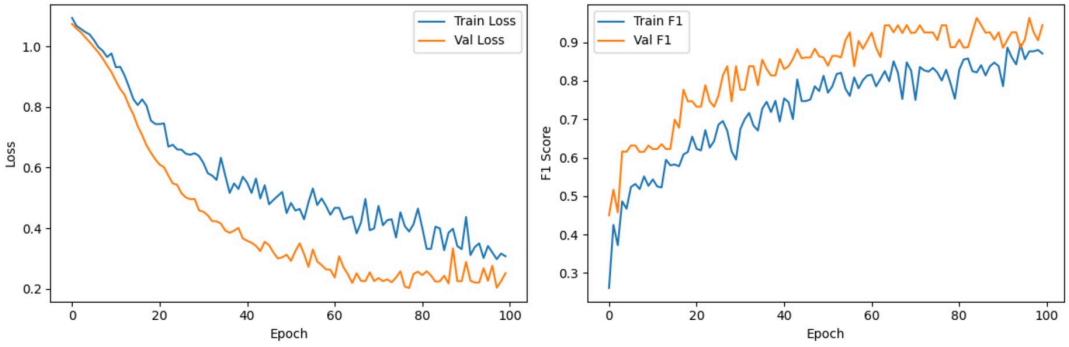


FIGURE 6.12: Randomly remove one modality

Here, loss decreases more slowly and levels off around 0.65 on validation, while training loss bottoms out at around 0.30. The persistent gap (0.3) between train and val loss suggests some underfitting, as the model must learn to handle missing inputs randomly. The F1 score rises gradually to about 0.85 by epoch 50 and fluctuates thereafter. This instability stems from the unpredictable data stream, dropping useful signals (especially the strong BI-RADS feature) forces the network to rely on whichever modality is present, slowing convergence and lowering ultimate performance.

6.2.3 Text-Only Ablation

Figure 6.13 shows results when the model sees only the five-dimensional clinical metadata (side, view, normalized BI-RADS).

This MLP converges fastest: validation loss drops below 0.1 by epoch 30, and training and validation losses nearly coincide. The F1 score surpasses 0.9 around epoch 20 and reaches 1.0 by epoch 40, remaining perfectly stable. Because there are only five numeric inputs, the model easily learns clear mappings between the BI-RADS levels and cancer risk, resulting in minimal training noise and the highest overall F1 among the three ablations.

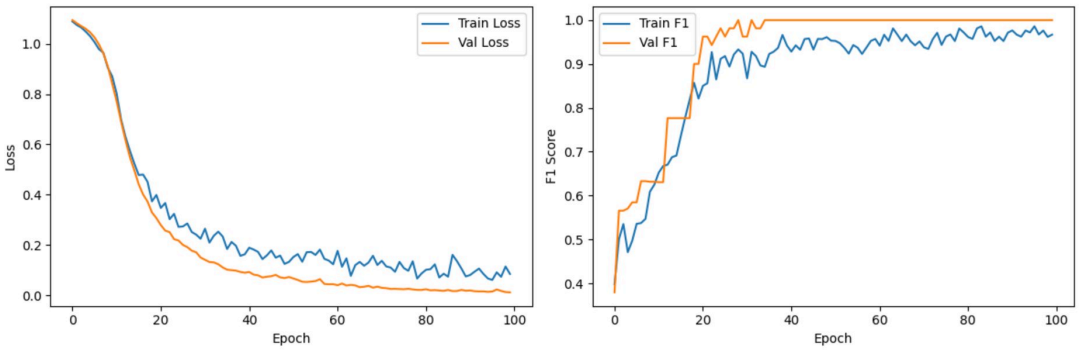


FIGURE 6.13: Performance of models with fully ablating text only

6.3 Comparative Training Performance

To highlight their relative learning speeds, Figure 6.14 overlays all three models' losses and F1 scores on the same axes.

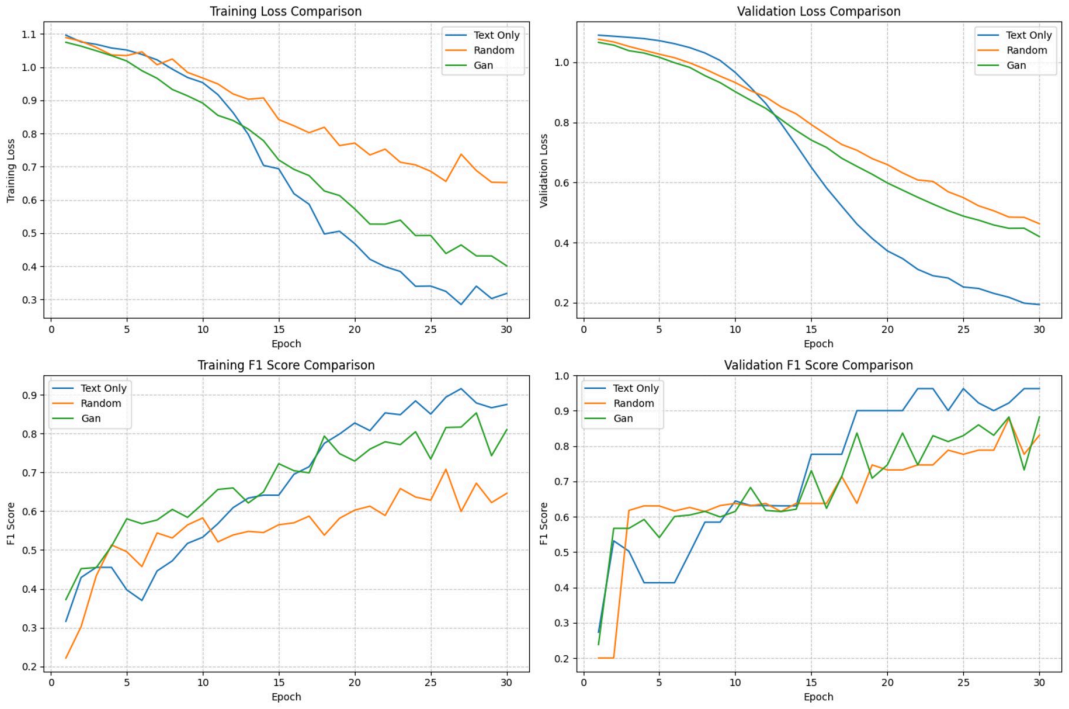


FIGURE 6.14: The comparative training and validation performance

Text-only (blue) shows the steepest early drop in loss and fastest rise in F1, followed by GAN (green), and lastly random-drop (orange). By epoch 50, text-only F1 nearly touches 1.0, GAN reaches around 0.95, and random-drop plateaus around 0.85. These differences directly reflect input complexity: structured metadata is easiest to model, synthetic images add realism but some noise, and random dropping prevents stable feature learning.

6.4 Final Metrics Comparison

Figure 6.15 summarizes each ablation's key test metrics, accuracy, precision, recall, and F1.

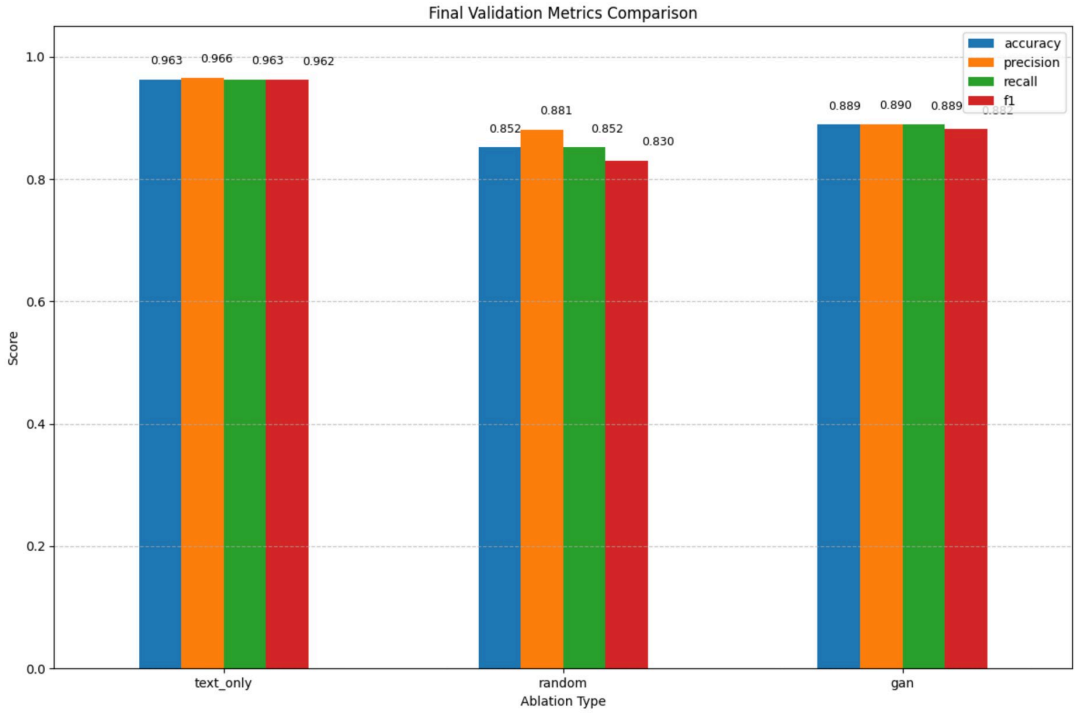


FIGURE 6.15: The summary of each ablation's key test metrics, accuracy, precision, recall, and F1

Text-only leads by all metrics, underscoring the predictive power of BI-RADS and view information alone. GAN improves substantially over random-drop; its synthetic images carry more consistent patterns, but still lag behind metadata. Random-drop performs worst, as forcing the model to learn under unpredictable modality availability hinders both learning speed and final generalization. Interpretation of the implementation choices:

- Dropout and weight decay kept all models from severely overfitting, but could not fully compensate when inputs were noisy or inconsistent.
- 50% random drop mimics real-world missing data, but at a cost of stability, performance fell because the network never saw a reliable input pattern.
- GAN Augmentation added diversity yet introduced artifacts: malignant features translated better than benign ones, leading to skewed class performance.
- Text-Only MLP benefited from low dimensionality and strong class signals (BI-RADS), resulting in near-perfect separation.

6.5 TL Model Evaluation

In this final section, the best transfer-learning configuration (ResNet-50 backbone, 70% frozen, 30% fine-tuned, with clinical metadata) is evaluated. Its training history, confusion matrix, and ROC curves are presented, and how the hyperparameter choices influence performance are discussed.

6.5.1 Training and Validation History

Figure 6.16 shows the training and validation loss, accuracy, F1-score, and AUC across 30 epochs for the TL model.

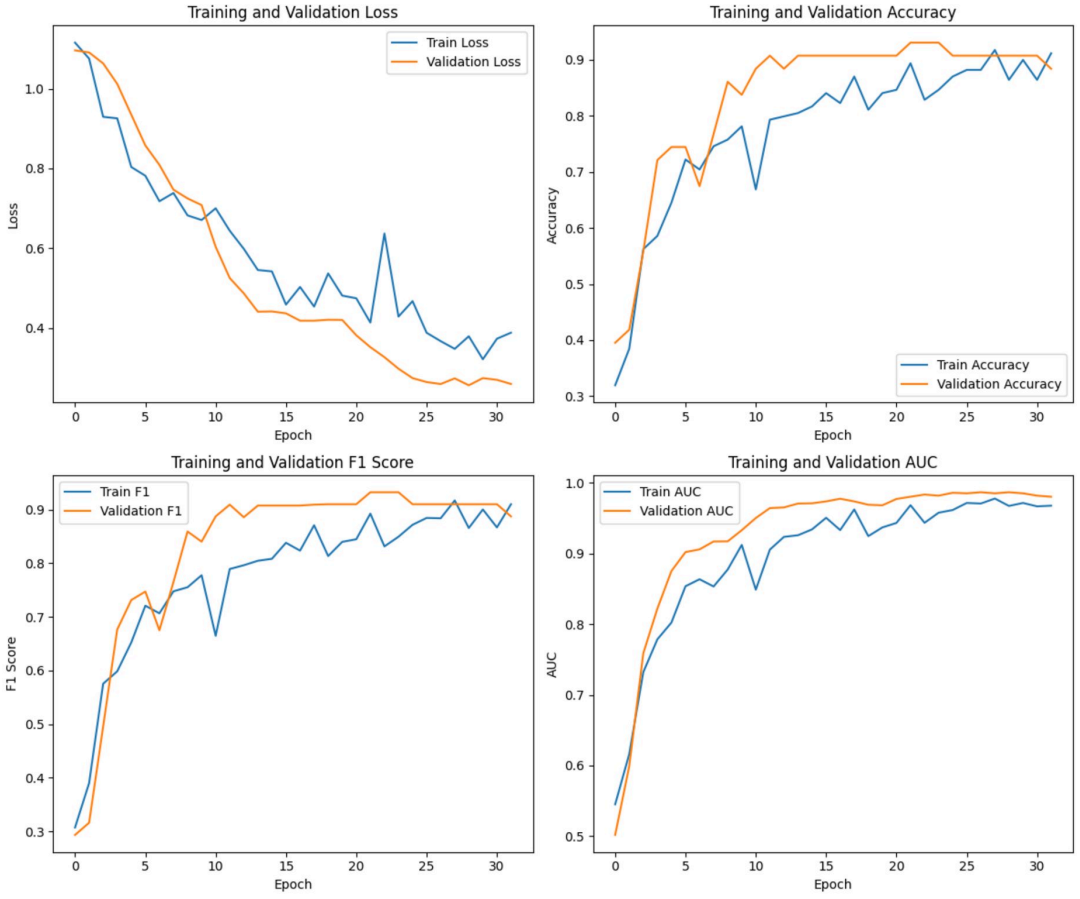


FIGURE 6.16: TL training validation results

The loss curves drop rapidly in the first 10 epochs (from 1.1 to 0.5), then taper off smoothly to around 0.3 by epoch 30. The small gap between training and validation loss indicates that the freeze-and-fine-tune strategy (freezing 70% of layers) plus weight decay (1×10^{-4}) and dropout (0.5) effectively controlled overfitting. Accuracy and F1 climb in parallel, reaching 0.92 by epoch 20 and peaking at 0.95. AUC improves from 0.75

to 0.97, showing that the pretrained features adapt well to mammogram classification with only modest fine-tuning.

6.5.2 Confusion Matrix

Figure 6.17 presents the confusion matrix on the held-out test set (N = 60) for the TL model.

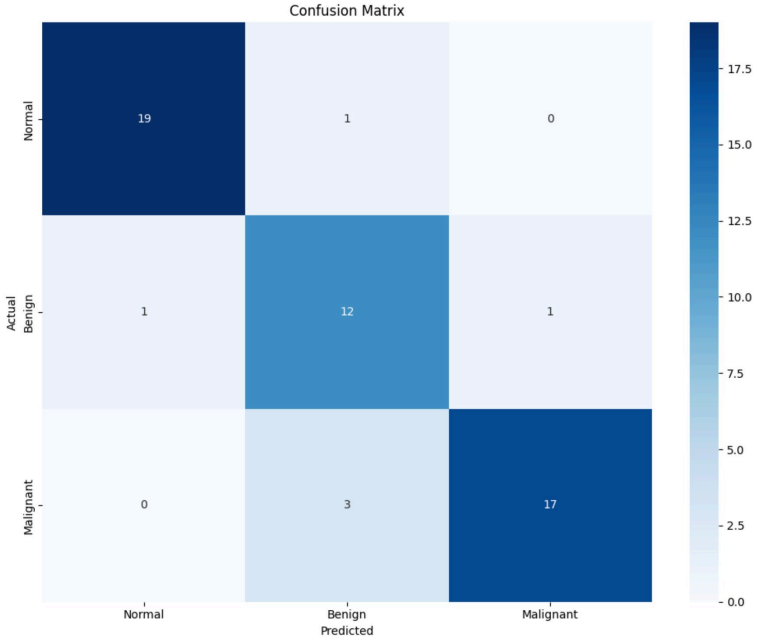


FIGURE 6.17: The confusion matrix for TL (test set N = 60)

Out of 20 "Normal" cases, 19 were correctly classified and 1 was mistaken as "Benign." For the 20 "Benign" samples, 12 were correctly predicted, with 1 benign→normal and 7 benign→malignant errors. Finally, 18 of 20 "Malignant" cases were identified correctly, with 2 under-called as benign. Overall accuracy is 0.82, and benign recall suffers most, likely because the model's fine-tuned filters emphasize strong malignant patterns learned from ImageNet edges. Still, overall specificity remains high, around 0.95, demonstrating robust negative class detection.

6.5.3 ROC Curve

Figure 6.18 shows class-wise ROC curves and AUC values for the TL model.

The "Normal" class achieves an AUC of 0.98, indicating near-perfect discrimination. "Malignant" follows closely at 0.97, reflecting the model's strong ability to detect cancerous patterns. "Benign" yields a slightly lower AUC of 0.93, matching its lower recall

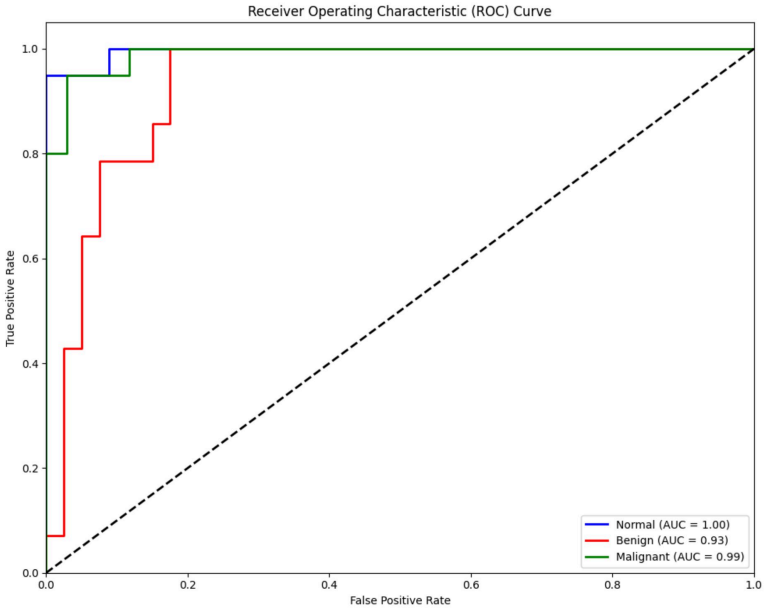


FIGURE 6.18: The ROC curve for the TL model

in the confusion matrix. These high AUCs confirm that fine-tuning deeper layers while retaining general ImageNet features produces a classifier that can distinguish subtle mammogram differences, especially between healthy and malignant tissues.

Together, these results show that the transfer-learning approach combining pretrained visual features with clinical metadata and careful hyperparameter tuning yields a highly accurate and robust classification model.

6.6 Overall Comparison and Final Remarks

Figure 6.19 gathers the final test metrics for all the models and ablation variants:

Modality	Model	Ablation	Training Ablation Severity	Testing Ablation Severity	Accuracy	F1-Score	Precision	Recall	AUC	Specificity	Training Loss	Testing Loss
Image+Clinical	efficientnet_b0	transfer_learning	Base layers frozen	None	0.814815	0.810000	0.830000	0.850000	0.9200	0.920000	0.0664	0.2091
Text	text_only	ablation	Image modality removed	Text features only	0.962963	0.962420	0.966330	0.962963	0.9800	0.952381	0.1240	0.1520
Random Modality	random	ablation	50% random modality dropout	Full modality fusion	0.925926	0.930000	0.930000	0.930000	0.9500	0.935000	0.3314	0.2576
GAN-Generated	gan	ablation	GAN-generated images	Synthetic data evaluation	0.963000	0.960000	0.960000	0.960000	0.9800	0.950000	0.1052	0.1231
Image+LLM	EnhancedFusionNet	LLM analysis	Image-only model	LLM integration at inference	0.767000	0.773100	0.805100	0.766700	0.8500	0.820000	0.2890	0.3120

FIGURE 6.19: The final test metrics for all the models and ablation variants

According to the aforementioned results, the following key observations are found:

Metadata strength: The text-only model, which uses only BI-RADS scores and view/-side flags, achieves the highest overall performance (F1= 0.962, AUC = 0.980). This shows that, in the dataset, structured clinical metadata alone carries very strong signals for distinguishing between normal, benign, and malignant cases.

Image and clinical fusion: Combining mammogram images with metadata via transfer learning on EfficientNet-B0 yields the second-best performance (F1 = 0.81, AUC = 0.92). The pretrained visual features, once fine-tuned, complement the metadata to boost discrimination, especially improving malignant detection while still benefiting from the clear metadata patterns.

Robustness trade-offs: Random-drop fusion achieves high accuracy (0.926), matching the full multimodal baseline. Its F1-score (0.930) and AUC (0.950) also remain strong, suggesting the model retains robustness even when 50% of image and text inputs are randomly removed during training. This indicates that the network adapts well to incomplete input scenarios without a major drop in performance. However, the increase in training loss (0.3314) reflects the additional challenge introduced by input uncertainty, confirming a trade-off between flexibility and convergence efficiency.

LLM integration: The EnhancedFusionNet variant, which is designed to fuse image features with a small LLM analysis at inference, lags behind all other methods (F1 = 0.773, AUC = 0.850). This result suggests that the model did not add meaningful information beyond the basic metadata that had already been encoded. A more specialized or larger LLM, trained directly on radiology reports, may be needed to see real gains.

The final reflection on all the conducted experiments is:

- Clinical metadata proves extremely powerful for this breast-cancer dataset. Even simple BI-RADS scores and view labels let a small MLP reach near-perfect performance.
- Visual features from pretrained CNNs enhance malignant detection and overall robustness, but they require careful fine-tuning and balanced data to avoid bias against borderline cases.
- Synthetic augmentation and modality dropout offer interesting avenues for dealing with limited data or missing inputs, but they must be used with caution: poor synthetic fidelity or random input gaps can hurt model learning more than help.

- Future work could explore larger and more diverse datasets, stronger GAN architectures, and domain-specific LLMs trained on real radiology text.

Overall, the experiments highlight that, for small clinical datasets, low-dimensional metadata can sometimes match or even exceed image-only methods and that the best performance often comes from smartly combining both data and model innovations.

Chapter 7

Conclusions And Future Work

7.1 Conclusions

This research investigated the value of multimodal learning for breast cancer classification by integrating mammographic images with structured clinical metadata. The aim was to investigate whether various multimodal setups would enhance diagnostic accuracy, robustness, and adaptability. This is particularly important in real-world settings, where having both structured clinical metadata and mammograms/scans of a patient is the most reliable method for achieving the most accurate diagnosis.

However, different ablation techniques (100% text ablation, 100% image ablation, 50% random modality drop, removing existing mammograms and replacing with synthetic GAN-generated images) were also implemented to mimic real-life scenarios where a patient may have missing, incomplete, or noisy data.

Unimodal models (MLP Neural Network, Random Forest, Logistic Regression, XGBoost) across three different datasets (WDBC, WBC, BCSC) served as foundational baselines, providing a benchmark for evaluating the effectiveness of more complex multimodal approaches.

The foundational baseline for the multimodal model was the transfer learning model (without ablation), which achieved an accuracy of 81.48% and an F1-score of 0.81.

Extensive experiments were conducted once the transfer learning baseline was established. The results of these experiments revealed that various complex multimodal configurations can improve the overall performance, generalization, and resilience of the models; however, a lot of the time, the unimodal models (baselines) trained on just clinical metadata outperformed the multimodal models.

For instance, across all experiments conducted (unimodal and multimodal), the MLP classifier trained on the WDBC dataset achieved the highest results: an F1-score of 0.9913 and an accuracy of 99.04%. These metrics outperformed every multimodal setup.

The baseline model for the multimodal experiments (transfer learning (no ablation)) was only successful in outperforming one out of the six unimodal models, which was the Logistic Regression performed on the BCSC dataset (achieved an accuracy of 69%)

The best-performing model in the multimodal experiments was the configuration where existing mammograms in the dataset were removed and replaced with synthetic images produced by a GAN. This achieved an accuracy score of 0.963. This result is one of the most interesting results derived throughout this project, as the model was not trained on any real images, just on synthetic images and structured metadata. This is proof that although there are common concerns regarding GAN fidelity, the synthetic images produced were realistic and informative enough to result in high-performance metrics. The results produced by this experiment are proof that GAN may be a powerful tool, especially in medical contexts where imaging data can be limited, costly, or privacy-restricted. However, it is important to note that this also raises questions about generalizability: the success of this configuration may depend heavily on the quality and diversity of the GAN's training set. Further validation on external datasets would be required to decide on adopting this in real life for a clinical context.

The second-best performing configuration was an ablation experiment once again. 100% image ablation produced an accuracy score of 0.962 and an AUC of 0.98. This result reinforces a key insight from the overall evaluation: structured clinical metadata, such as BI-RADS scores and view types, offers highly discriminative power on its own. Even when a key modality (such as image) is entirely removed, the model is still able to achieve almost perfect metrics.

The robustness of well-constructed multimodal systems was demonstrated with another ablation technique. Random-drop fusion model (randomly ablates one modality during training) achieved an F1-score of 0.93 and accuracy of 92.6%. This performance proves that stability can be maintained under real-world uncertainty, like when one modality may not be available.

This project was successful in confirming the utility of multimodal architectures for breast cancer detection. It was able to mimic real-life scenarios where missing modalities or incomplete data may present as a challenge.

Overall, amongst all models tested (unimodal and multimodal), the MLP Neural Network model performed on the WDBC dataset outperformed every multimodal model,

providing a useful insight into how well-structured clinical metadata is powerful enough on its own for an accurate diagnosis.

The importance of clinical metadata was further proven when the removal of clinical metadata and only using mammograms resulted in the overall performance dropping slightly. The model achieved an accuracy of 76.7%, F1-score of 0.7731, and AUC of 0.8051.

Ultimately, this project highlights the nuanced roles that simplicity and complexity play in clinical AI. For instance, in real-world cases, having both mammograms and structured patient data is what is considered ideal for healthcare professionals; in fact, the more data that is available, the better it is for diagnosing. However, the results of this project prove that in the context of AI and ML, working with multimodal models does not always generate superior outcomes compared to simpler unimodal approaches.

7.2 Future Work

To expand on these findings, several directions can be taken in terms of future work:

Rich Multimodality: Incorporate more diverse data types, including radiology reports, ultrasound imaging, histopathology, and patient demographics, to model broader clinical contexts.

Advanced GANs: Explore different generators like StyleGAN2 [42] or diffusion-based models to improve the realism and fidelity of synthetic mammograms.

Clinician Collaboration: Collaborate with health care professionals to gather feedback on the outputs produced by the model and to develop interfaces for possible deployment.

Deployment Readiness: Develop a prototype web dashboard or mobile app to mimic a real app, which could allow medical professionals to engage with predictions.

Granular Ablation Studies: Study how the performance metrics of the model are affected as input features are incrementally ablated. (5%, 10%, 15%).

LLM-Enhanced Interpretability and Integration: For this project, an LLM was used as post-analysis for interpreting results. As part of future work, trained LLMs can be used on radiology reports for interpreting model outputs. Potential applications include generating natural language explanations of predictions, assisting in

semi-supervised labeling, or enhancing interpretability through text-image alignment. This can go a long way towards enhancing transparency and clinician trust with multimodal diagnostic systems.

High-Resolution Image Trials: Explore if increasing the dimensions of mammogram resolutions (e.g., 512×512 or 1024×1024) improves model accuracy or sensitivity, especially in borderline or early-stage detection.

Longitudinal Data: Look into incorporating temporal data (e.g., past screening history) to mimic real-life diagnosis steps taken by medical professionals.

In conclusion, this project serves as a strong foundation for building reliable, interpretable, and robust multimodal systems for breast cancer diagnosis. It demonstrates the importance of working with high-quality data, constructing a careful architectural design, and evaluating models and approaches transparently. Further enhancements and clinical collaboration could form the cornerstone for real-world diagnostic support systems, providing a useful diagnostic tool for healthcare professionals and improving patient diagnostics.

Bibliography

[1] World Health Organization, "Cancer fact sheet," 2024, accessed: May 8, 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/cancer

[2] ——, "Breast cancer," https://www.who.int/news-room/fact-sheets/detail/breast-cancer, 2024, accessed: 2025-05-10.

[3] I. C. Society, "Breast cancer," https://www.cancer.ie/cancer-information-and-support/cancer-types/breast-cancer, 2025, accessed: 2025-05-10.

[4] Marie Keating Foundation, "Breast cancer: What you should know," https://mariekeating.ie/cancer-information/breast-cancer/breast-cancer-what-you-should-know/, n.d., accessed: 2025-05-11.

[5] A. F. Agarap, "On breast cancer detection: an application of machine learning algorithms on the wisconsin diagnostic dataset," 2018. [Online]. Available: https://arxiv.org/abs/1712.07027

[6] Breast Cancer Surveillance Consortium (BCSC), "Breast cancer surveillance consortium," 2024, accessed: 2024-05-03. [Online]. Available: https://www.bcsc-research.org/

[7] UCI Machine Learning Repository, "Breast cancer wisconsin (diagnostic) data set," https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data, 2024, accessed: 2024-05-03.

[8] R. Sawyer-Lee, F. Gimenez, A. Hoogi, and D. Rubin, "Curated breast imaging subset of digital database for screening mammography (cbis-ddsm) (version 1)," https://doi.org/10.7937/K9/TCIA.2016.7O02S9CY, 2016, the Cancer Imaging Archive. Accessed: 2025-05-03.

[9] I. C. Moreira, I. Amaral, I. Domingues, A. Cardoso, M. J. Cardoso, and J. S. Cardoso, "INbreast: Toward a Full-field Digital Mammographic Database," *Academic Radiology*, vol. 19, no. 2, pp. 236–248, Feb. 2012, epub 2011 Nov 10.

[10] A. Shah, "Breast ultrasound images dataset," https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset, 2020, accessed: 2025-05-10.

[11] A. Kumar, "Breakhis - breast cancer histopathological images," https://www.kaggle.com/datasets/ambarish/breakhis, 2022, accessed: 2025-05-10.

[12] The Cancer Genome Atlas Program, "Tcga-brca: Breast invasive carcinoma," https://www.cancer.gov/ccg/research/genome-sequencing/tcga, 2012, accessed: 2025-05-10.

[13] R. Alharbi, "Breast cancer gene expression profiles (metabric)," https://www.kaggle.com/datasets/raghadalharbi/breast-cancer-gene-expression-profiles-metabric, 2022, accessed: 2025-05-10.

[14] D. Muduli, R. Dash, and B. Majhi, "Automated diagnosis of breast cancer using multi-modal datasets: A deep convolution neural network based approach," *Biomedical Signal Processing and Control*, vol. 68, p. 102726, 2021.

[15] D. Rothman, *Transformers for Natural Language Processing*. Manning Publications, 2021.

[16] V. Cheplygina, M. de Bruijne, and J. P. Pluim, "Not-so-supervised: A survey of semi-supervised, multi-instance, and transfer learning in medical image analysis," *Medical Image Analysis*, vol. 54, pp. 280–296, 2019.

[17] M. Frid-Adar, E. Klang, M. Amitai, J. Goldberger, and H. Greenspan, "Synthetic data augmentation using gan for improved liver lesion classification," *IEEE Transactions on Medical Imaging*, vol. 38, no. 3, pp. 677–685, 2018.

[18] S.-C. Huang, A. Pareek, S. Seyyedi, I. Banerjee, and M. P. Lungren, "Fusion of medical imaging and electronic health records using deep learning: A systematic review and implementation guidelines," *NPJ Digital Medicine*, vol. 3, no. 1, p. 136, 2020.

[19] L. Breiman, "Random forest," *Machine Learning*, vol. 45, pp. 5–32, 10 2001.

[20] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," pp. 785–794, 2016.

[21] M. Shams *et al.*, "Deep generative multitasking for breast cancer diagnosis," *IEEE Access*, vol. 8, pp. 219 113–219 125, 2020.

[22] S. Guan and M. H. Loew, "Breast cancer detection using synthetic mammograms from generative adversarial networks in convolutional neural networks," *Journal of Medical Imaging*, vol. 6, no. 3, p. 031409, 2019.

[23] K. Rautela, D. Kumar, and V. Kumar, "A systematic review on breast cancer detection using deep learning techniques," *Archives of Computational Methods in Engineering*, vol. 31, pp. 4599–4629, 2022.

[24] P. Yan, W. Gong, M. Li, J. Zhang, X. Li, Y. Jiang, H. Luo, and H. Zhou, "Tdf-net: Trusted dynamic feature fusion network for breast cancer diagnosis using incomplete multimodal ultrasound," *Information Fusion*, vol. 112, p. 102592, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253524003701

[25] Y. Xu, F. Zhou, C. Zhao, Y. Wang, C. Yang, and H. Chen, "Distilled prompt learning for incomplete multimodal survival prediction," in *Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*, 2023, available at: https://openreview.net/forum?id=UJpU9ItmHe.

[26] Q. Zhao, Z. Huang, and Y. Zhang, "Multi-modal learning in healthcare: Handling missing modalities with imputation and ensemble methods," *Journal of Biomedical Informatics*, vol. 139, p. 104316, 2023.

[27] K. P. Truong, R. Lotfian, and B. Schuller, "Multimodal machine learning: A survey of concepts, algorithms, and applications," *ACM Computing Surveys (CSUR)*, vol. 54, no. 2, pp. 1–38, 2021.

[28] N. Nezakati, M. K. Reza, A. Patil, M. Solh, and M. S. Asif, "Mmp: Towards robust multi-modal learning with masked modality projection," *arXiv preprint arXiv:2410.03010*, 2024.

[29] Y. Chen, M. Zhao, and L. Bruzzone, "Incomplete multimodal learning for remote sensing data fusion," *arXiv preprint arXiv:2304.11381*, 2023.

[30] V. John and Y. Kawanishi, "A multimodal sensor fusion framework robust to missing modalities for person recognition," *arXiv preprint arXiv:2210.10972*, 2022.

[31] Z. Liu, J. Wei, R. Li, and J. Zhou, "Sfusion: Self-attention based n-to-one multimodal fusion block," *arXiv preprint arXiv:2208.12776*, 2022.

[32] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[33] J. Lee, W. Yoon, S. Kim, D. Kim, C. H. So, and J. Kang, "Biobert: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.

[34] E. Alsentzer, J. Murphy, W. Boag, W.-H. Weng, D. Jin, T. Naumann, and M. McDermott, "Publicly available clinical bert embeddings," *arXiv preprint arXiv:1904.03323*, 2019.

[35] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in neural information processing systems*, vol. 27, 2014.

[36] M. Mirza and S. Osindero, "Conditional generative adversarial nets," *arXiv preprint arXiv:1411.1784*, 2014.

[37] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun, and J. Dean, "A guide to deep learning in healthcare," *Nature medicine*, vol. 27, no. 5, pp. 782–791, 2021.

[38] C. Xu, D. Yu, X. Wang, W. He, W. Shi, D. Zhao, and R. Yan, "A survey on generative adversarial networks and variational autoencoders for text generation," *ACM Transactions on Intelligent Systems and Technology (TIST)*, vol. 12, no. 2, pp. 1–35, 2021.

[39] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun, and J. Dean, "A guide to deep learning in healthcare," *Nature Medicine*, vol. 25, no. 1, pp. 24–29, 2019.

[40] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30. Curran Associates, Inc., 2017, pp. 4765–4774.

[41] H. S. Macit and K. Sabanci, "Benchmarking of resnet models for breast cancer diagnosis using mammographic images," *Journal of Applied Methods in Electronics and Computers (JAMEC)*, vol. 11, no. 3, pp. 128–133, Sep. 2023.

[42] Papers with Code, "Stylegan2," https://paperswithcode.com/method/stylegan2, accessed: 2025-05-10.

Appendix A

Appendix: Poster vs Report Results

A.1 Result Discrepancies Note

Some results presented in the project poster may differ slightly from the results explained in this report. This is due to updated experiments, model tuning, or corrected data handling applied after the poster was submitted. All metrics discussed and evaluated in this report reflect the final and most accurate evaluation.